

Module 15

Investment Decisions

Learning Objectives

By the end of this module you should understand:

1. The three steps in the investment process
2. The concept of present value and its application in investment appraisal:
 - Discounted cash flow
 - Net present value
 - Internal rate of return
3. When to use these three techniques in evaluating projects
4. The impact of risk, uncertainty and inflation in the investment decision, particularly the application of sensitivity analysis
5. How to calculate a company's cost of capital for discounting purposes.

Agenda

15.1 Introduction

15.2 The Investment Process

15.3 Concept of Present Value

15.4 Discounted Cash Flow Approach

15.5 Net Present Value (NPV)

15.6 Discounted Cash Flow (DCF) Rate of Return

15.7 Comparison of Net Present Value and DCF Rate of Return

15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations

15.9 Risk and Uncertainty and Inflation

15.10 Risk and Uncertainty

15.11 Payoff or Payback Period

15.12 Sensitivity Analysis

15.13 Risk Analysis

15.14 The Key Investment Factors

15.15 Projected Average Cost of Capital

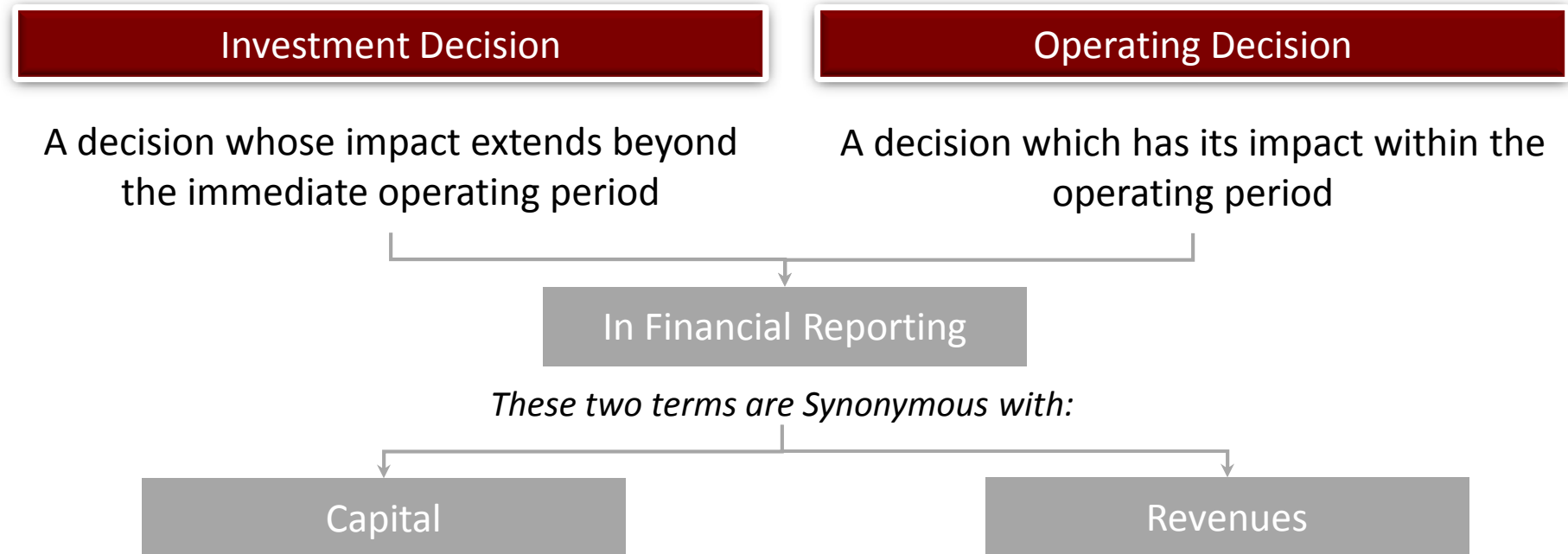
15.16 Average Cost of Capital

15.17 Opportunity Cost, Risk and the Cost of Capital

15.18 Investment Appraisal and Inflation

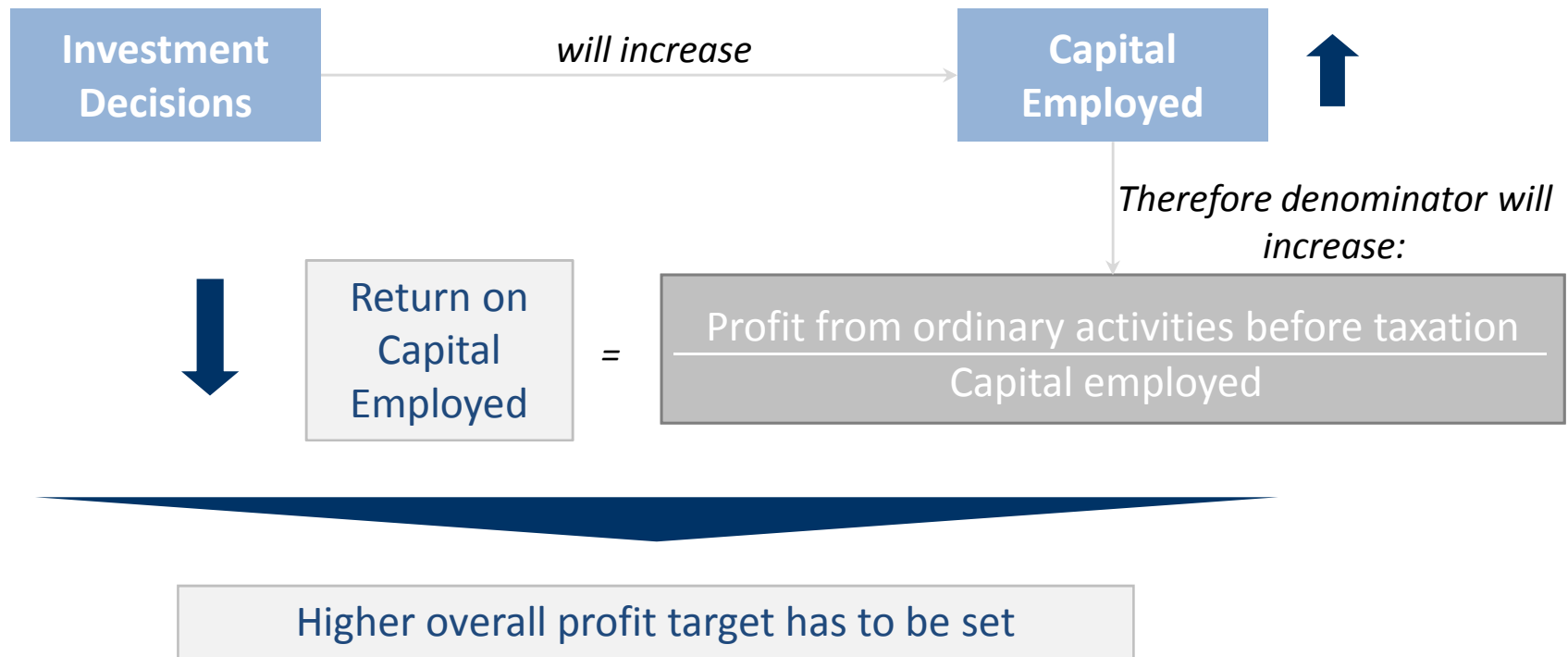
15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

There are two types of decisions facing management: investment and operating; the investment decision is more important



This module is primarily concerned with the financial aspects of the investment decisions underlying capital expenditures

There's a direct relation between investment decisions and the Return on Capital Employed (ROCE)



Capital projects often involve large sums of money, therefore the financial consequences of a 'bad' or 'wrong' decision can be severe

Agenda

15.1 Introduction

15.2 The Investment Process

15.3 Concept of Present Value

15.4 Discounted Cash Flow Approach

15.5 Net Present Value (NPV)

15.6 Discounted Cash Flow (DCF) Rate of Return

15.7 Comparison of Net Present Value and DCF Rate of Return

15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations

15.9 Risk and Uncertainty and Inflation

15.10 Risk and Uncertainty

15.11 Payoff or Payback Period

15.12 Sensitivity Analysis

15.13 Risk Analysis

15.14 The Key Investment Factors

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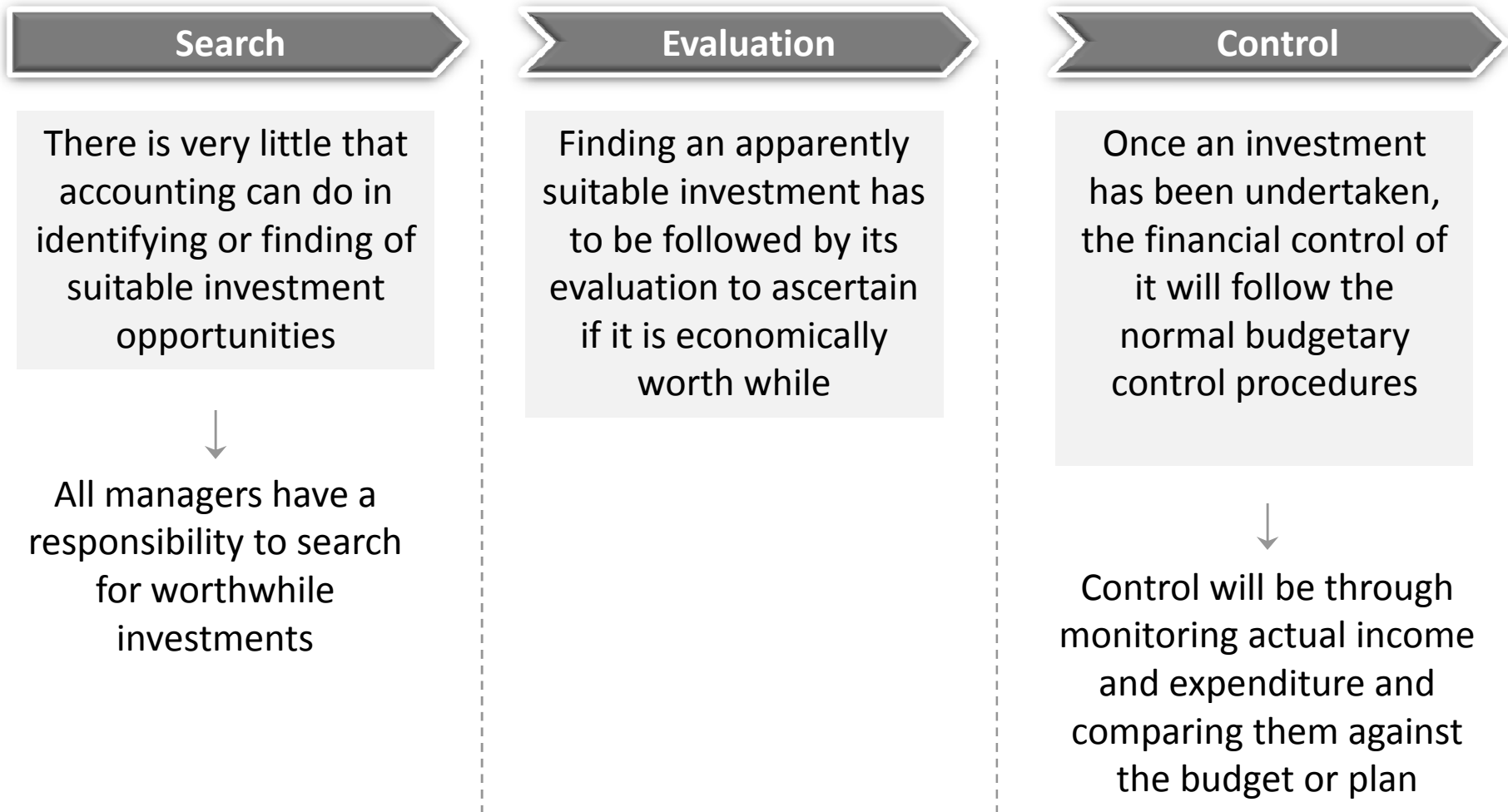
15.16 Average Cost of Capital

15.17 Opportunity Cost, Risk and the Cost of Capital

15.18 Investment Appraisal and Inflation

15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

The investment process involves three quite distinct, although related, processes – search, evaluation and control

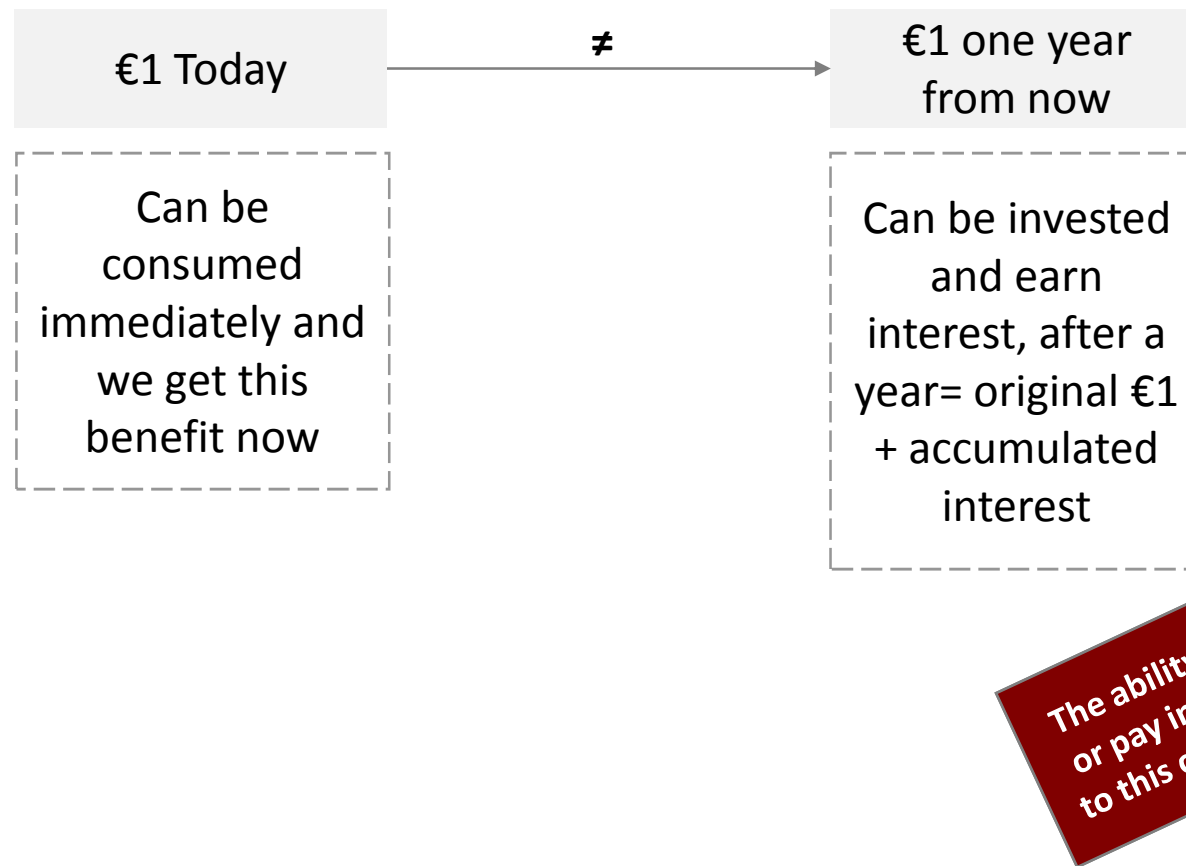


The next section is primarily about the framework within which evaluation takes place

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

Because of investment decisions' impact on future time periods, we need a concept that takes into consideration the time value of money



That's why the concept uses an interest rate to try to give an equivalent value to monies payable or receivable at different points of time

Note that the concept of present value is different than that of inflation and risk

Inflation

€1 after a year will not represent
the same purchasing power as
today's

Risk

€1 after a year will not represent
the same €1 today
Today is safer because you have
it in your pocket now and you
don't know what a year will
bring

Even in an inflation- and risk-free world there is a time value of money and this is our immediate concern

This concept can best be illustrated through this simple example

Example 1

- If **€100** is invested for one year at annual interest rate of **10%**=> it will amount to **€110**
- The original investment of **€100** plus the **€10** interest

Example 2

- If the **€100** is invested for two years => it will amount to **€121**
- The original **€100** plus the first year's interest of **€10** amounting to **€110** plus the second year's interest of **€11** (10% of **€110**) added on to this to give the total amount of **€121**

This can be simplified as follows:

€100 invested for 2 years at 10% will amount to $€100 \times 1.10 \times 1.10 = €121$.

This is the original sum $\times (1 + \text{the rate of interest}) \times (1 + \text{the rate of interest})$ which equals the original sum $\times (1 + \text{the rate of interest})^2$.

Then we come up with this general rule for future value of money and present value of money

If:

n

The numbers of years

i

The rate of interest

Then:

**Future
Value**

Any sum invested for n years at i per cent will amount to that sum multiplied by $(1 + i)^n$

€100 invested for two years at 10% amounts to $€100 \times (1 + 0.10)^2$ which equals €121. This €121 can be referred to as the future value

We can use the reciprocal of the equation to calculate the Present Value

$$\frac{1}{(1 + i)^n}$$

A sum of €121 receivable in two years at 10% has a present value today of only €100

$$€121 \times \frac{1}{(1 + 0.10)^2}$$

Present value is the sum which would have to be invested today to amount to a given sum at a rate of interest over a given time period

Exercise

The present value today of €200 in two years at 9% is:

$$\frac{200}{(1 + 0.09)^2} = \text{€}168.4$$

The present value today of €170 in one year at 9% is:

$$\frac{170}{(1 + 0.09)^1} = \text{€}155.9$$

The present value today of €300 in ten years at 9% is:

$$\frac{300}{(1 + 0.09)^{10}} = \text{€}126.6$$

These different values of €200, €170 and €300 receivable or payable at different future times can now be ranked as they have now been reduced to present or equivalent values

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

The present value concept forms the basis of what is popularly known as the discounted cash flow approach to the evaluation of investment opportunities

Discounted Cash Flow Approach

We **discount the cash flows** arising out of the investment down to a present or equivalent value



So that we can:

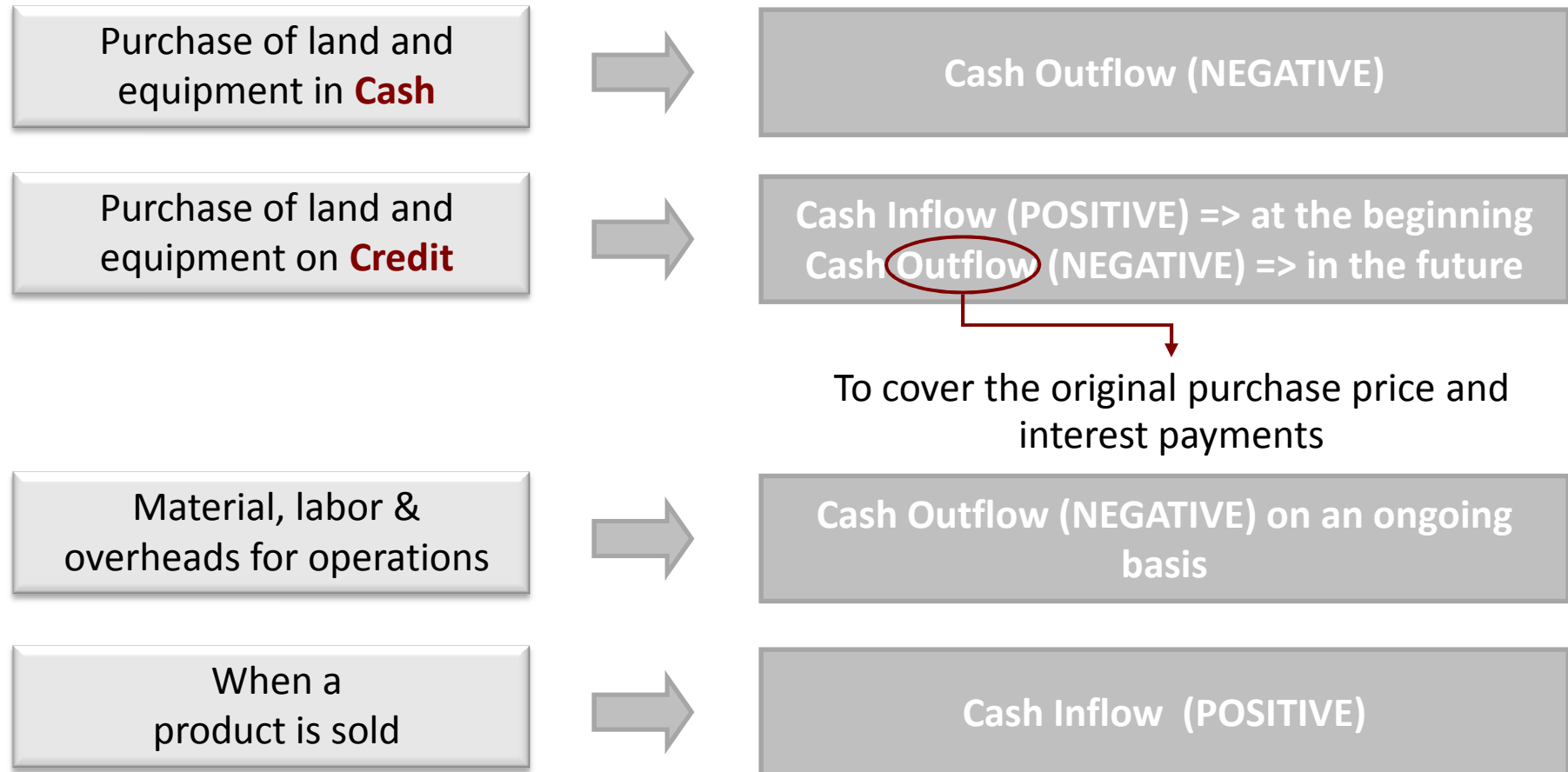
Calculate the profitability and/or cost of it

Through:

Net Present Value (NPV)

Internal Rate of Return

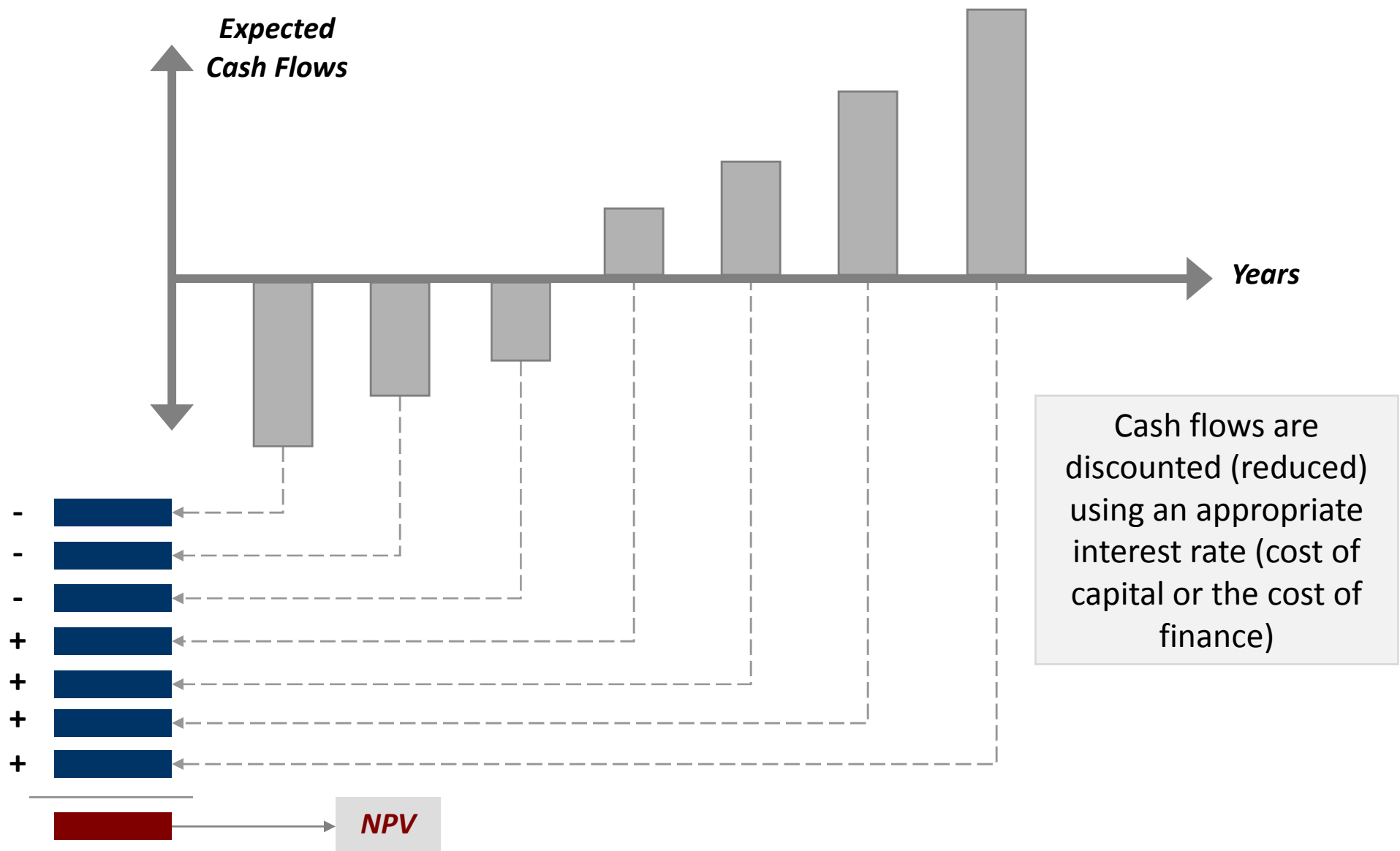
Cash flows are either negative or positive



Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

The net present value approach takes all the cash flows associated with a project and discounts them to a common denominator (PV) using an appropriate interest rate



For projects where the manager is faced with a simple accept or reject situation the decision rule is quite simple using NPV

IF:

The NPV > 0	➔	The rate of return is greater than the cost of financing the project	<i>ACCEPT</i>
NPV < 0	➔	The rate of return is lower than the cost of financing the project	<i>REJECT</i>
NPV = 0	➔	The rate of return is equal to the cost of financing and the project breaks even	<i>RECONSIDER</i>

One of the difficulties associated with net present value is that it is not a concept that enters into everyday business conversation, so not all managers understand it

We will illustrate how difficult it is to translate net present value into the language of management through the following simple example

Example

Suppose an investment of €300 now will produce cash flows of €114 for the next three years and that the cost of capital is 5%. The cash profile of the investment is as follows:

	Cash flow
Year 0 (beginning of the investment)	–€300
Year 1	114
Year 2	114
Year 3	114
Unadjusted cash flow	<u>€ 42</u>

We first conduct our NPV analysis

The net present value of this positive cash flow at 5% is:

$$-\text{€}300 + \frac{\text{€}114}{(1.05)^1} + \frac{\text{€}114}{(1.05)^2} + \frac{\text{€}114}{(1.05)^3} = \text{€}10.5$$

If we set the present values alongside the cash flows we can see the impact of adjusting the cash flows by the 5% cost of capital

	Cash flow	Net present value at 5% of cash flow
Year 0	–€300	–€300
1	114	108.50
2	114	103.40
3	114	98.60
	€ 42	€ 10.50

From this analysis we can immediately deduce a number of observations which are of interest to the manager

1

Project is profitable because it has a positive present value

2

It has a rate of return greater than 5%

3

If the project is undertaken and these results achieved, then the wealth of the business will increase by, in today's terms, €10.50

4

If there is a choice between this project and another one which gives a smaller increase in wealth, that is, has a lower present value, then this is the one to choose => Increase Shareholders' wealth by €10.50

We'll now consider the situation if we are to borrow this amount at a 5% cost of financing; here we have to build a specific table

- If we knew with absolute certainty that the investment of €300 would generate the cash flows of €114 per annum for three years and that the cost of financing this was 5% then we could spend/consume €10.50 now.
- Even if we had to borrow all of the €310.50, that is, the €300 for the investment and the €10.50 to spend now, at the end of the three years table which shows the repayment schedule on the assumption that we borrow €310.50 at 5% and use €300 for the investment and consume the other €10.50 by, say, spending it on a holiday or some other pleasure, and then use the annual cash flow of €114 from the investment to pay off the loan of €310.50.

Year	Capital due at beginning	5% interest	Total due	Cash flow (repayment)	Capital due at end
1	€310.50	€15.50	€326.00	€114	€212.00
2	212.00	10.60	222.60	114	108.60
3	108.60	05.40	114.00	114	–

To calculate the net present value therefore we need to estimate all the cash flows – capital and revenue, positive and negative

Let's study another illustration

- The AB Company is considering the investment of €100,000 in capital equipment. The equipment will last for ten years at the end of which it will have a scrap value of €2,500.
- The net operating cash flows (cash income received less cash expenses paid out) will be €27,500 per annum for the first three years and thereafter €37,500 per annum. The company's cost of capital is 15%.

First

Establish the cash flows in the relevant time periods

Second

We then find the present value of these at 15%

These two steps can be found illustrated in the following table

		Cash flow	15% PV factor	PV at 15%
		€		€
Year	0 Capital investment	–100 000	1.00	–100 000
	1 Operating cash flow	27 500	0.870	23 925
	2 Operating cash flow	27 500	0.756	20 790
	3 Operating cash flow	27 500	0.658	18 095
	4 Operating cash flow	37 500	0.572	21 450
	5 Operating cash flow	37 500	0.497	18 637
	6 Operating cash flow	37 500	0.432	16 200
	7 Operating cash flow	37 500	0.376	14 100
	8 Operating cash flow	37 500	0.327	12 262
	9 Operating cash flow	37 500	0.284	10 650
	10 Operating cash flow	37 500	0.247	9 263
	11 Scrap value	2 500	0.215	537
Net present value				65 909

It shows whether a project is profitable and whether the rate of return is above or below the cost of capital, but it does not tell us what the actual rate of return is

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

The DCF return technique has some similar requirements to that of the NPV approach

Similar Requirements

- Applies the present value concept
- A profile of the cash flows

New Requirements

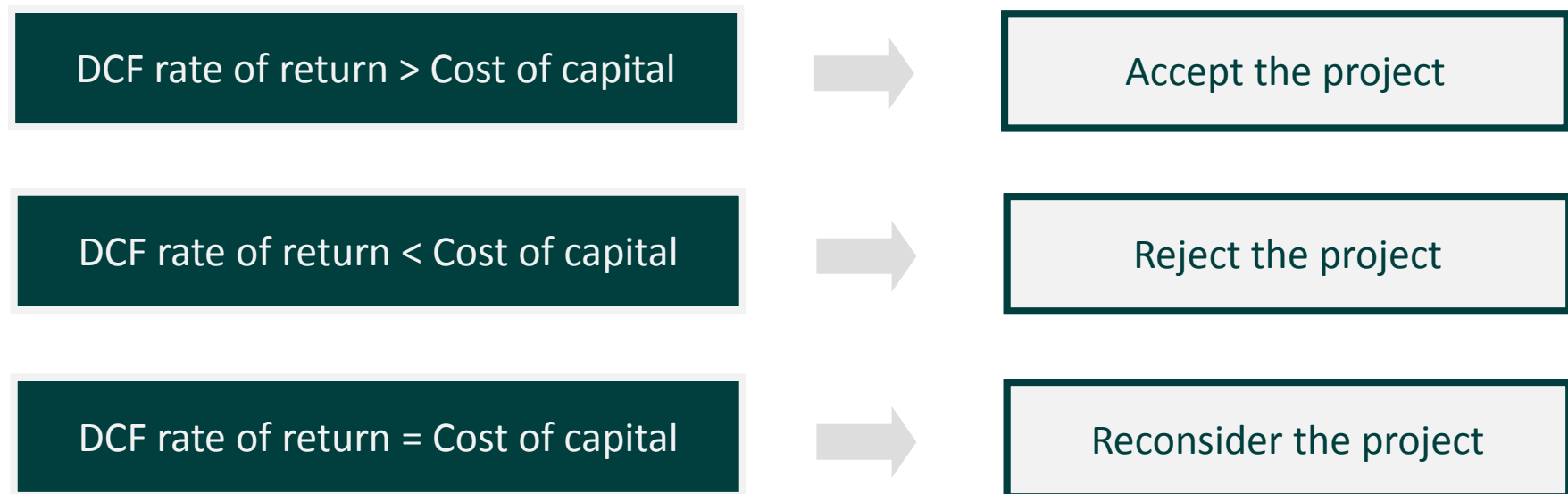
- No discounting by the cost of capital
- Instead we find the interest rate which reduces all these cash flows to zero



The interest rate (rate of return) at which the project will exactly break even

This is the maximum rate which a business could pay on borrowed money and still not make a loss and therefore will be compared with the cost of borrowings, that is, the cost of capital to see if the project is worth undertaking

The financial decision rule for the DCF rate of return is expressed in fairly simple terms



It must be emphasized that these are financial decision rules. Projects by themselves are rarely decided solely on financial grounds

We'll use our previous example of the investment of €300 and the annual cash inflows of €114 for three years to get the DCF rate of return

The DCF rate of return is 7%

First Illustration to the solution:

$$-€300 + \frac{€114}{(1.07)^1} + \frac{€114}{(1.07)^2} + \frac{€114}{(1.07)^3} = 0$$

An alternative presentation of the solution can be shown here:

		Capital outstanding at commencement of period	7% interest	Total out- standing	Cash flow	Capital outstanding at close of period
Year	1	€300	21	321	114	207
	2	207	14	221	114	107
	3	107	7	114	114	0

(NB: Figures have been rounded to nearest whole number.)

Provided the cost of capital to the business is less than 7%, this project is profitable

This can be illustrated again by using the example of the AB Company, previously considered

		€
Year	0	–100 000
	1	27 500
	2	27 500
	3	27 500
	4	37 500
	5	37 500
	6	37 500
	7	37 500
	8	37 500
	9	37 500
	10	37 500
	11	2 500

The interest rate which will give all of the above cash flows a total present value of zero is just over 28%. The project therefore has a rate of return of just over 28%

The DCF rate of return is more popular with managers

It answers the following question:

‘What return on capital will we get if we invest in new plant?’

The actual calculation of the DCF rate of return can be arithmetically tedious

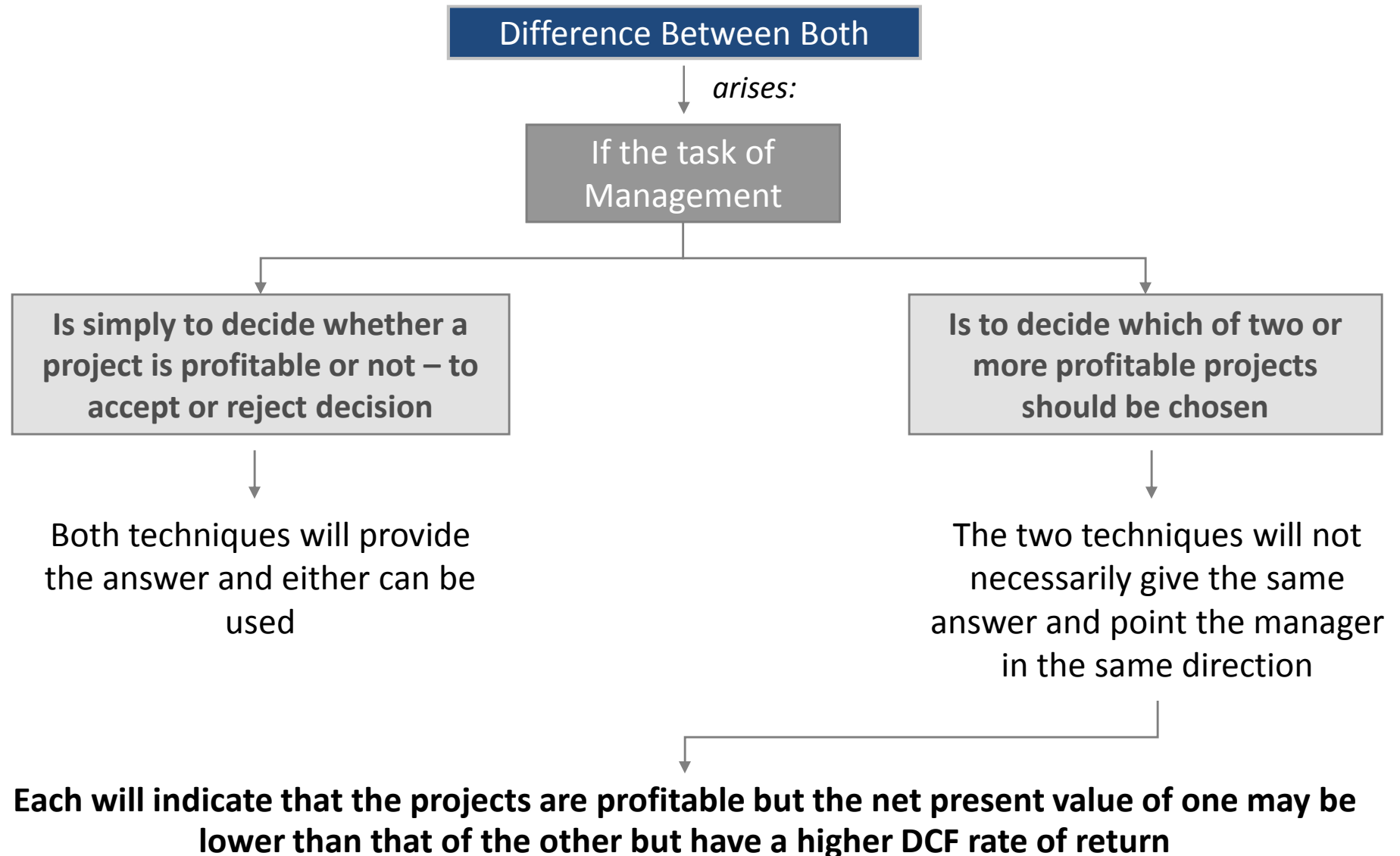


All computers and semi-sophisticated calculators are or can be programmed to do this calculation + step by step explanation in Appendix 15.2

Agenda

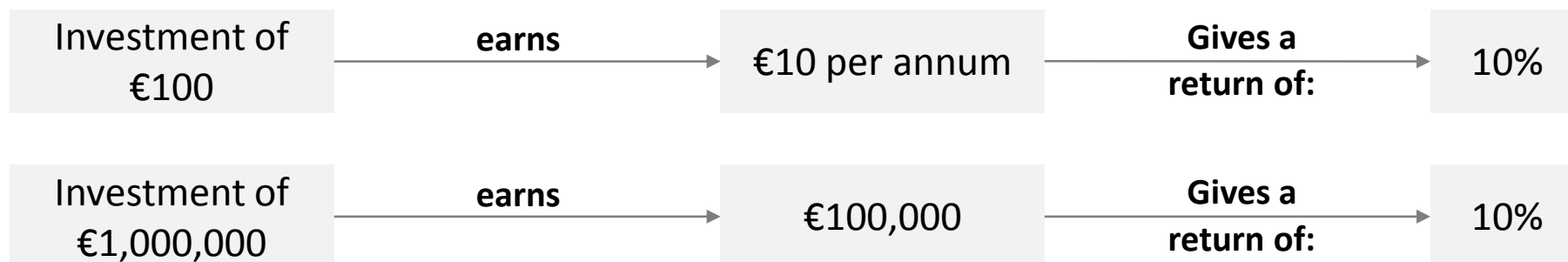
- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

Both NPV & DCF rate of return are based on the discounted cash flow principle & measure profitability considering the time factor & the recovery of the investment



NPV is an absolute measure whereas the rate of return is a relative measure

Relative Measure

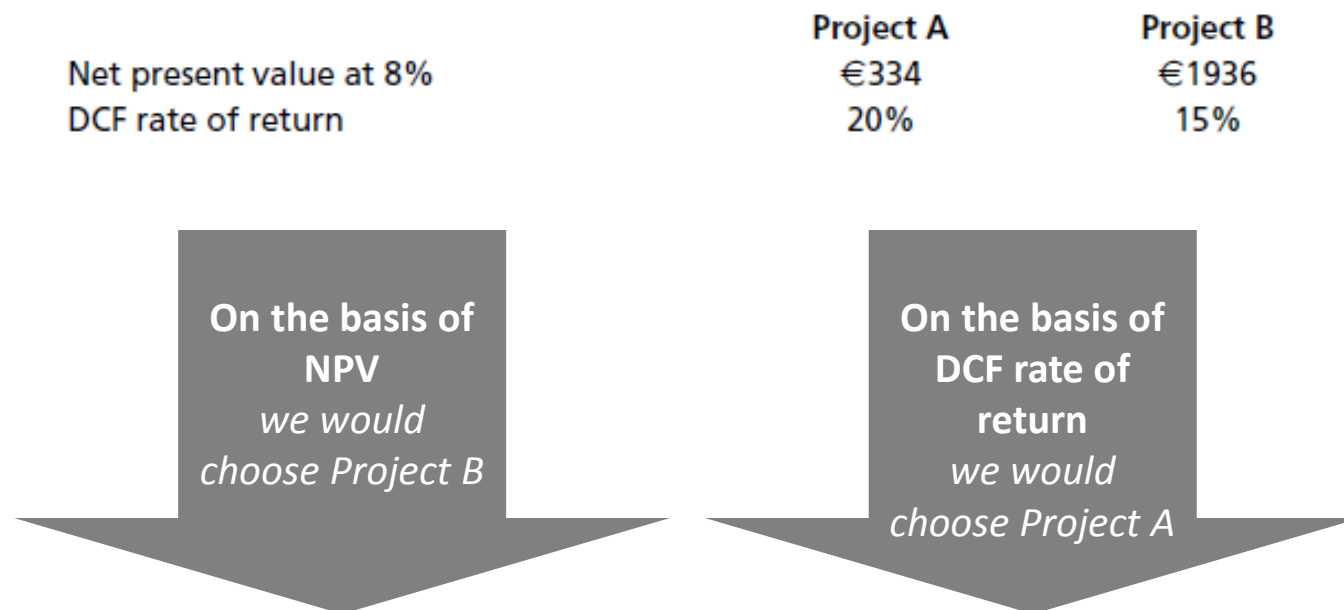


This is the relative measure; that is, it relates the return to the original investment

This is unlike NPV where the present values of these will differ quite significantly because of the different magnitudes of the sums involved

Example

An organization is considering two mutually exclusive proposals. Both require the same piece of land so that if one project is selected the other has to be abandoned. Project A requires an investment of €2,000 and generates cash flows of €1,309 per annum for two years whereas Project B requires an investment of €20,000 and generates cash flows of €12,302 per annum for two years. If we apply the appraisal techniques using 8% as the cost of capital we get the following:



So which one do we choose?

Management accounting techniques always require the exercise of judgment and in this case further analysis can be done which will help in this judgment

In this case we can take the difference between the projects and analyze them as follows:

	Project A cash flow €	Project B cash flow €	Difference B – A €
Year 0	–2 000	–20 000	–18 000
1	1 309	12 302	10 993
2	1 309	12 302	10 993

Project B gives us all that Project A gives us plus a return of €10,993 per annum for two years for an investment of €18,000

If we accept Project A, we will save the initial investment of €18,000 but will have foregone €10,993 per annum for two years

What could we do with the extra €18,000 which would be available if we accepted A in favor of B? If, for example, the next best project available required an investment of €18,000 and only generated cash flows over two years of €10,000 per annum

We could see that it would be better in the first place to go for Project B because this is better than the combination of Project A and the next best available project

It is also useful to apply the normal appraisal analysis on this difference

This is the DCF rate of return on an investment of €18,000 with cash flows of €10,993 per annum for two years, and its NPV is at 8%

	Cash flow of difference between projects A and B €
Year 0	–18 000
1	10 993
2	10 993
NPV @ 8%	1 602
DCF rate of return	14%

We can now see that Project B gives the same rate of return as Project A on the same level of investment plus 14% on the extra



In all cases the first step in evaluation is to calculate the profitability of an investment opportunity. This is the key information most often required by managers

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

There are many cases where capital budgeting involves a cost-only situation in for-profit and not-for-profit organizations

For-Profit Organizations:

Example 1

The construction of a new building which is not expected to generate additional revenue

Example 2

The installation of a central heating system or the acquisition of new equipment for a staff dining room



The approach used:

The net present value and apply the **corollary of the profit maximization** objective



Which is the:

Cost Minimization Objective

Cost minimization occurs where the net present value is lowest; management will want to know the costs involved, including the cost of rejecting the cheapest

Consider the following example

Suppose an organization is faced with installing a central heating system in a new building and has the choice of two different systems or it could be two tenders for the same system. One system has a higher capital cost but lower operating costs than the other. The organization's cost of capital is 16%. If the different systems have cash flows over their respective lives of six years as follows then the evaluation will be:

		System 1			System 2		
		Cash flows	Discount factor	PV at 16%	Cash flows	Discount factor	PV at 16%
Year	0	−€8 000	1.000	€8 000	−€12 000	1.000	€12 000
	1	−2 000	0.862	1 724	−1 500	0.862	1 293
	2	−2 000	0.743	1 486	−1 500	0.743	1 114
	3	−2 500	0.641	1 602	−1 500	0.641	961
	4	−2 500	0.552	1 380	−2 000	0.552	1 104
	5	−3 000	0.476	1 428	−2 000	0.476	952
	6	−3 000	0.410	1 230	−2 000	0.410	820
Net present value				<u>€16 850</u>			<u>€18 244</u>

System 1 is the cheaper; that is, it has the lower present value, €16,850 compared to €18,244

On a cost minimization basis System 1 would be the better

Here again if it helps, we can evaluate the difference between the two systems to give an additional indication of the financial superiority of one over the other

	Cash flows system 1	Cash flows system 2	Difference	
Year 0	−8 000	−12 000	−4 000	
1	−2 000	− 1 500	500	
2	−2 000	− 1 500	500	
3	−2 500	− 1 500	1 000	
4	−2 500	− 2 000	500	
5	−3 000	− 2 000	1 000	
6	−3 000	− 2 000	1 000	
NPV at 16%			−1 394	<i>Not only does System 1 have a lower present value by €1,394, but also the investment of the additional €4,000 to achieve savings in costs of €500 in Years 1, 2, and 4, and €1,000 in Years 3, 5, and 6, gives a return of only 3%</i>

The approach of evaluating differences is a practical way of incorporating into the management accounting system a very important concept for management, that of opportunity cost

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

The discounted cash flow approach to investment appraisal is based on a number of assumptions

Zero Risk

Certainty

No Inflation

These assumptions which are fairly unrealistic in the real world, reinforce the point that in using any techniques management must always exercise judgment

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

The weakest assumption made to date was that of certainty (risk & uncertainty are synonymous in this context)

Risk/ Uncertainty

All of the factors which are likely to make an actual result differ from an estimate or budget



Approaches for tackling the problem of risk/ uncertainty

No further analysis at all

Estimates are prepared on a **'most realistic estimate in the circumstances'** basis and the evaluation is done using a discounted cash flow approach

Management then looks at the figures which this produces and decides what to do

'High-Hurdle' Test

Management groups projects into certain **risk categories**, e.g. high, medium or low, e.g. replacement of existing equipment is put into a different (usually lower risk) category than investment in completely new facilities; others may group according to geographical location

Once the risk categories have been established, the next step is to determine the hurdle that is applicable to each of them

Perhaps the most common ‘high hurdle’ is to require a higher rate of return for riskier projects

A three-range classification could be:

Classification	Target rate of return
Low risk	12%
Medium risk	16%
High risk	20%

While this undoubtedly imposes a more severe acceptance test on the higher risk projects, it does not distinguish between the individual components which together make up the complete project

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

Another 'high hurdle' is known as 'payoff' or 'payback' and is so popular that it is often given the status of an evaluation technique in its own right

Payback

The time taken for the positive cash flows to recoup the original investment



In a risk situation, payback tries to answer the question:

'When will we get our money back?'

Example showing how to calculate the payback period

An investment of €50,000 generates the following cash flows:

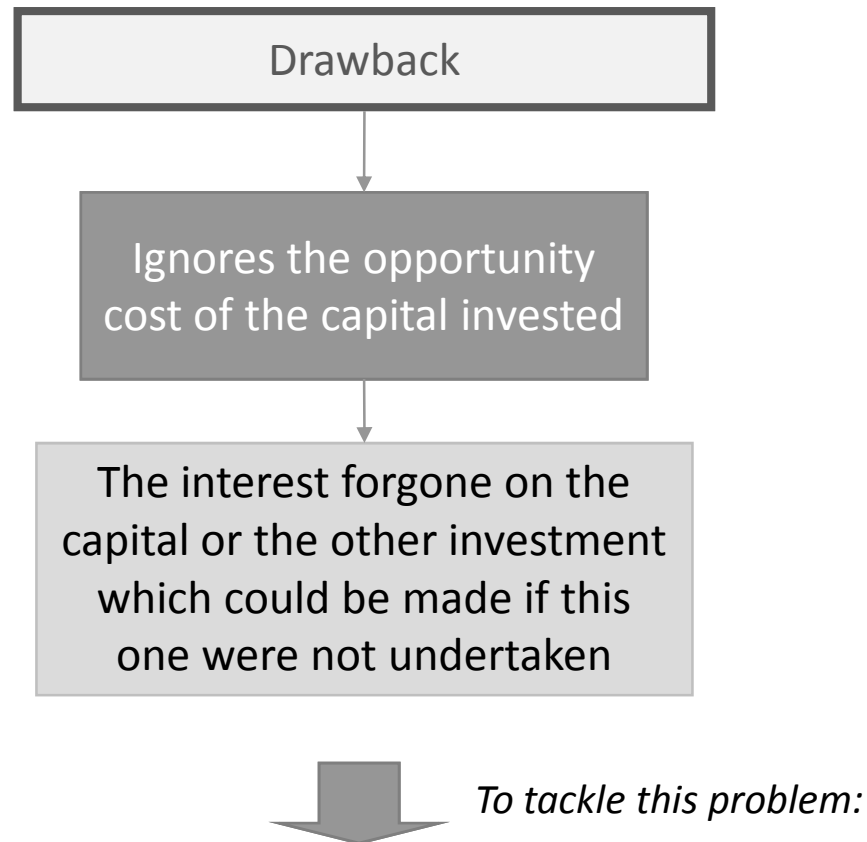
Year 1	€13 000
2	15 000
3	15 000
4	15 000
5	15 000
6	10 000

The cumulative cash flows will therefore be:

Year	1	– €37 000 ($-50\,000 + 13\,000$)
	2	– 22 000 ($-37\,000 + 15\,000$)
	3	– 7 000 ($-22\,000 + 15\,000$)
	4	+ 8 000 ($-7\,000 + 15\,000$)

The payoff or payback period therefore will be just under 3.5 years

But the payback period has a drawback which is that it ignores the opportunity cost of the capital invested



We can reduce the cash flows to their present value before deducting them from the original investment or add on to the outstanding capital an interest element based on the cost of capital

Discounting future cash flows before deducting it from the initial investment will give a higher payback period as shown below in the example

Using the same examples as before, and with a 10% cost of capital, the discounted payback can be calculated as follows:

		Cash flows	Discount factor	PV of cash flows	Cumulative discounted cash flows
Year	0	–€50 000	1.000	–50 000	–50 000
	1	+ 13 000	0.909	+11 817	–38 183
	2	+ 15 000	0.826	+12 390	–25 793
	3	+ 15 000	0.751	+11 265	–14 528
	4	+ 15 000	0.683	+10 245	– 4 283
	5	+ 15 000	0.621	+ 9 315	+ 5 032
	6	+ 10 000	0.564	+ 5 640	+10 672

The real payback period is now 4.5 years and not 3.5 years as previously calculated

The alternative method adds on to the outstanding capital an interest element based on the cost of capital

Add Interest (10%) to the outstanding capital and then deduct the undiscounted cash flows from this until the capital plus interest has been eliminated

Capital outstanding	Year 0	–50 000	
Interest 10%		5 000	
		–55 000	
Cash repaid	Year 1	13 000	
		–42 000	
Interest		4 200	
		–46 200	
Cash repaid	Year 2	15 000	
		–31 200	
Interest		3 120	
		–34 320	
Cash repaid	Year 3	15 000	
		–19 320	
Interest		1 932	
		–21 252	
Cash repaid	Year 4	15 000	
		– 6 252	
Interest		625	
		– 6 877	
Cash repaid	Year 5	15 000	(Capital + Interest repaid mid-way through Year 4)
		+ 8 123	

Both payback methods suffer from the same drawbacks; this makes them inadequate as stand-alone evaluation techniques

Payback:

1

Implies that there is a standard payback period

There need not be the same standard payback for all types of projects

2

Ignores one of the most important factors in investment, which is profit

Payback is a break-even position and thus profit only occurs after payback

e.g.

Classification	Target payback period
Low risk	5.0 years
Medium risk	3.5 years
High risk	2.5 years

Nevertheless the problem still remains of what the 'correct' payback period is and how you determine it

However, it is useful if it is used along with one of the more appropriate evaluation techniques such as present value or DCF rate of return

Although, there may be no ‘correct’ payback period, management use it to get an overall ‘feel’ for the project

Example

Suppose, for example, that there are two mutually exclusive projects, each concerned with heavy mining operations in the same place but on different scales. The evaluation might be as follows:

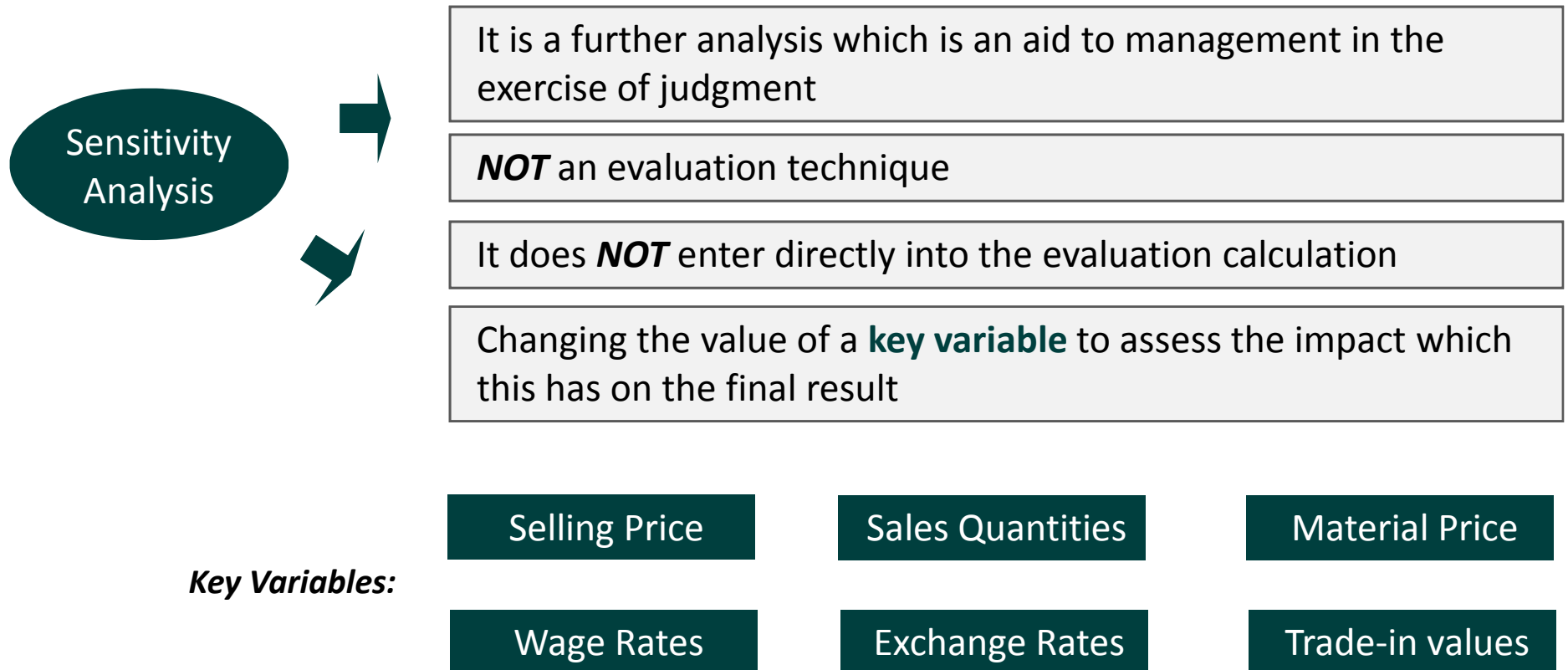
	Option A	Option B
Investment	€5 000 000	€20 000 000
NPV	€2 500 000	€11 000 000
Cost of capital	14%	14%
Rate of return	19%	24%
Payback	2.5 years	6 years

Despite the higher profitability of Option B management might still prefer to go for Option A because of, say, risk factors such as political instability and reliance on government concessions

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

Another approach for coping with risk and uncertainty is sensitivity analysis



Given the large number of variables involved in a real-life situation it is normally assumed that management is aware of the likely critical or key factors and will concentrate attention on these

We will now conduct a sensitivity analysis on the AB Company to test the variations concerning the selling price

Back to Example

	Cash flow	15% PV factor	PV at 15%
	€		€
Year 0 Capital investment	–100 000	1.00	–100 000
1 Operating cash flow	27 500	0.870	23 925
2 Operating cash flow	27 500	0.756	20 790
3 Operating cash flow	27 500	0.658	18 095
4 Operating cash flow	37 500	0.572	21 450
5 Operating cash flow	37 500	0.497	18 637
6 Operating cash flow	37 500	0.432	16 200
7 Operating cash flow	37 500	0.376	14 100
8 Operating cash flow	37 500	0.327	12 262
9 Operating cash flow	37 500	0.284	10 650
10 Operating cash flow	37 500	0.247	9 263
11 Scrap value	2 500	0.215	537
Net present value			65 909

We saw that the rate of return was just over 28% and with a cost of capital of 15% had a net present value of €65,909, and has a discounted payback in 5 years

Let's now see how we'll conduct the sensitivity analysis on that company

Suppose now that the forecasters are reasonably confident about their forecast of production/sales volumes, capital costs, wages and other operating costs but are uncertain about selling prices because of the relatively volatile market in which trading will take place.

Let us assume also that in setting the forecasts annual sales revenue was estimated as follows:

Years 1–3	€70 000 per annum
Years 4–10	€75 000 per annum

In this example we'll see the outcome if the selling prices were 10% higher or lower than forecast, assuming that all other variables are the same

In order to test the impact of variations in selling prices (a sensitivity analysis) we can recalculate what the position would be if selling prices were, say, $\pm 10\%$, that is, a sales revenue of €70,000 per annum would increase or decrease by €7,000

		Original cash flow	Cash flow with 10% decrease in selling prices	Cash flow with 10% increase in selling prices
Year	0	–€100 000	–€100 000	–€100 000
	1	27 500	20 500	34 500
	2	27 500	20 500	34 500
	3	27 500	20 500	34 500
	4	37 500	30 000	45 000
	5	37 500	30 000	45 000
	6	37 500	30 000	45 000
	7	37 500	30 000	45 000
	8	37 500	30 000	45 000
	9	37 500	30 000	45 000
	10	37 500	30 000	45 000
	11	2 500	2 500	2 500
NPV at 15%		65 909	29 409	102 410

10% decrease in prices shows a profit, but much lower than before

If selling prices were more than 18% lower than expected the project would run at a loss unless some of the other variables could be adjusted

Despite of its usefulness in highlighting the impact of a change in the forecast value of a variable, sensitivity analysis does have its drawbacks

Drawback

Does **NOT** give, nor build into the evaluation, any indication of the likelihood of such a variation taking place

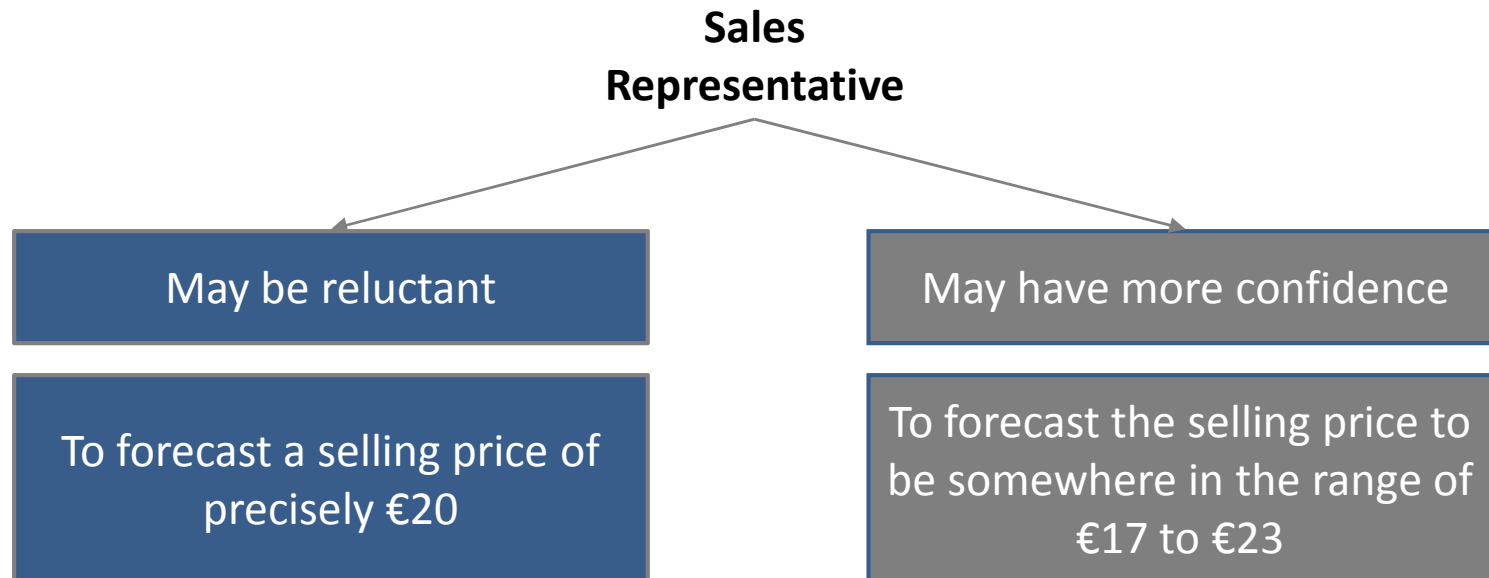
Profitability may be highly sensitive to variations in the cost of capital but the probability of a change in this variable may be very small

The consideration of this probability of change comes under what is called risk analysis

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

It is important to give some consideration to the likelihood of change when it is expected that it may take place



Risk analysis is about taking this idea of a range of likely values and applying the mathematical techniques of probability analysis to it in order to give a better ‘feel’ for the riskiness of a project

At the heart of risk analysis is the need to get a range of values with the likelihood of these occurring as opposed to the more common single-point estimate

Risk Analysis

Getting a range of values with the likelihood of these occurring as opposed to the more common single-point estimate

Then take this range and find the expected value (the mean of the probability distribution) and this is the value used in the appraisal

Example

Suppose that a sales representative estimates that there is a 25% chance that a selling price will be €10, a 50% chance that it will be €12 and a 25% chance that it will be €13. This could be expressed as follows:

Forecast selling price	Probability	Expected value
€10	0.25	€2.50
12	0.50	6.00
13	0.25	3.25
		<u>Most likely value €11.75</u>

Probabilities always add up to 1 →

1.00

Probability of 1

Means that it is certain to happen

Probability of 0

Means that it is certain not to happen

The selling prices used actually deviate from this expected value and the extent of this deviation can also give us a feel for the riskiness associated with the variable – the greater the deviation the greater the risk

Two factors may have the same expected value but a different risk

Example

Variable A		
Forecast value	Probability	Expected value
€10	0.25	€2.50
12	0.50	6.00
13	0.25	3.25
	<u>1.00</u>	<u>€11.75</u>

A is within the range €10–€13

Variable B		
Forecast value	Probability	Expected value
€ 6	0.05	€ 0.30
8	0.10	0.80
11	0.35	3.85
13	0.35	4.55
14	0.10	1.40
17	0.05	0.85
	<u>1.00</u>	<u>€11.75</u>

B has a much wider range, €6–€17

Therefore B could be regarded as having a greater risk attached to it than A

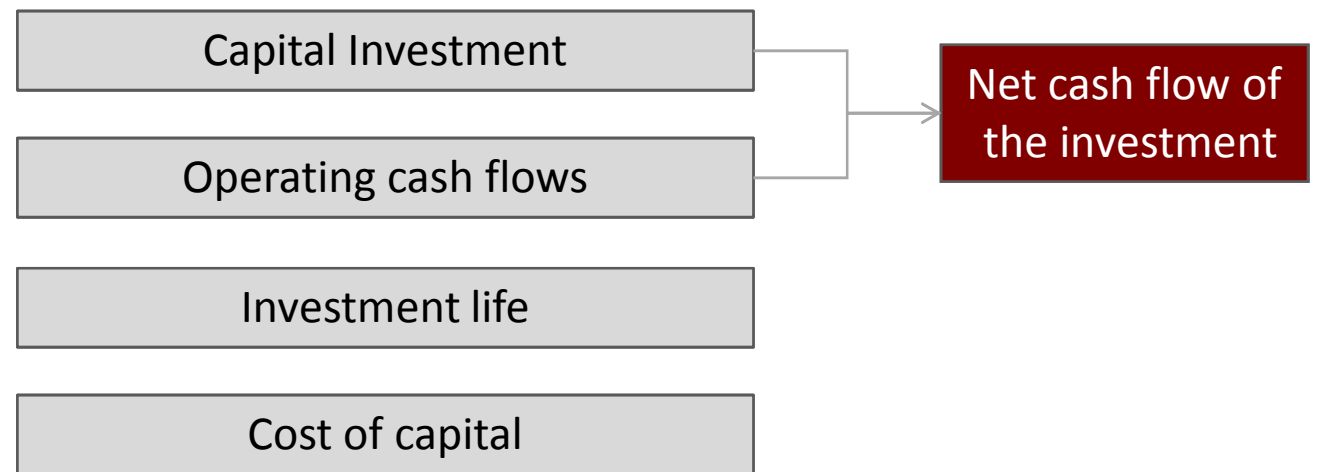
If, for example, both had the same unit cost, the chances of making profits or losses are greater with B than with A

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

Investment appraisal techniques highlight the key investment factors which are central to the profitability of an investment

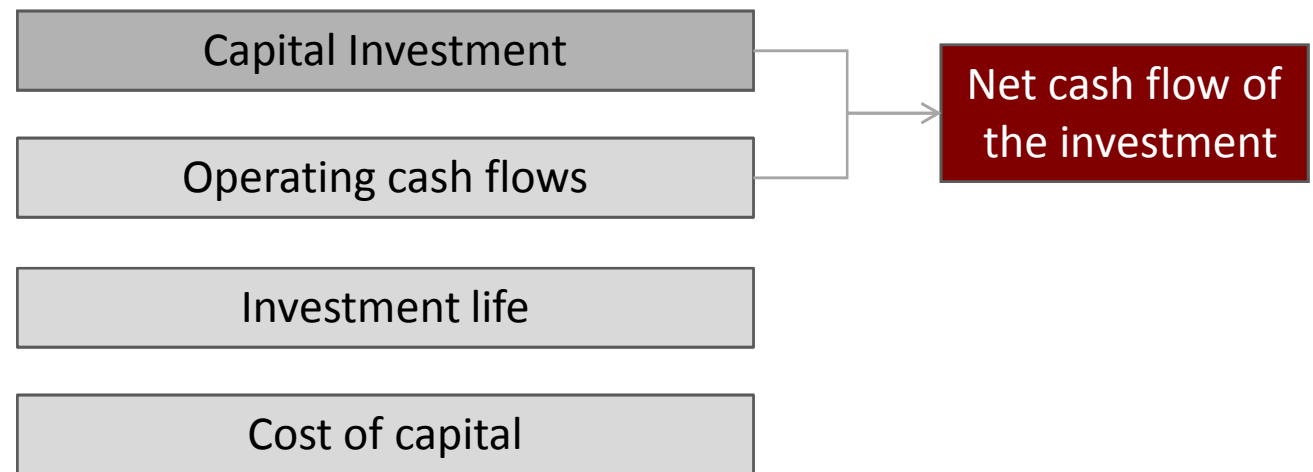
They are:



All four are interrelated and eventually have to be looked at in unison

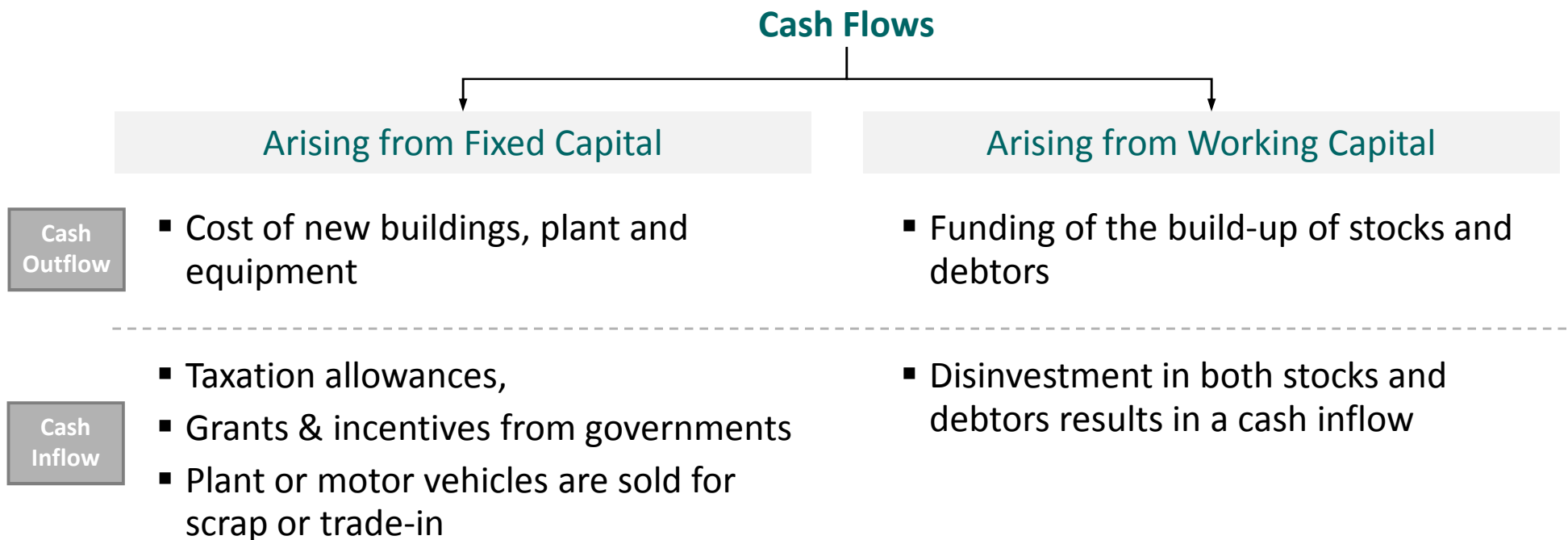
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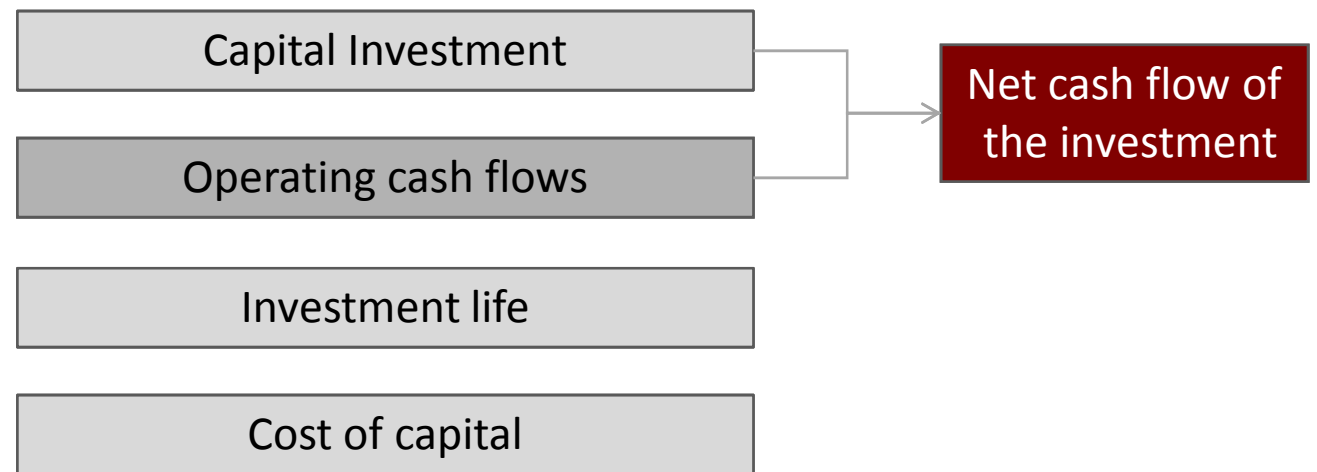
Capital Investment cash flows consist of the amount and timing of both total fixed and working capital that will be required for the specific project



It is important to allocate these cash flows to the appropriate time period; a critical factor is when a new plant will become operational & when products will be manufactured & sold

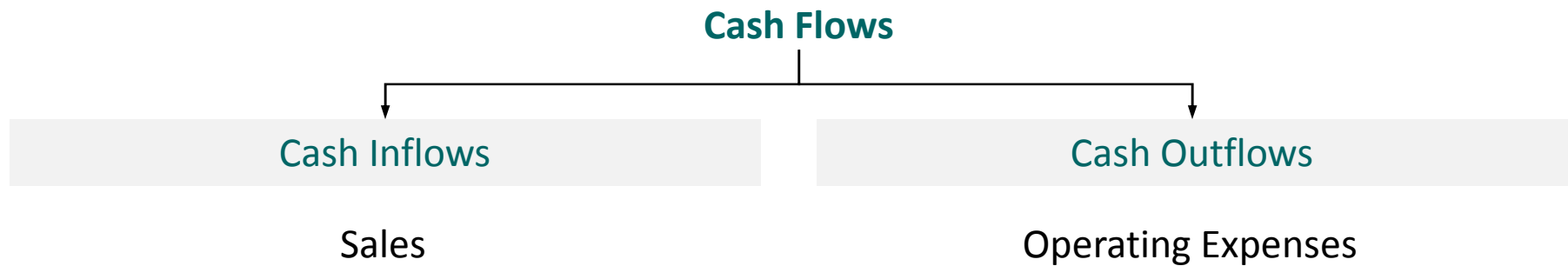
Investment appraisal techniques highlight the key investment factors which are central to the profitability of an investment

They are:



All four are interrelated and eventually have to be looked at in unison

Operating cash flows are concerned with the cash flows generated and with the operating period they occur in



Remember that non-cash expenses such as depreciation which are essentially accounting book entries have no part to play in these calculations

Investment appraisal techniques look at profitability over the total life of the project, whereas financial reporting tries to allocate profit to a specific accounting period

Example

Suppose an investment of €100,000 in new equipment generates operating cash flows of €40,000 per annum for five years with no recovery of the original investment. The DCF rate of return on this is just over 28.5% but if the equipment is written off equally over the five years – straight-line depreciation – then the published financial statements will be:

Year 1 Profit for year €20 000 (€40 000 less depreciation €20 000).

Balance sheet as at Year end 1

	Cost	Depreciation	Book value
Fixed asset	€100 000	€20 000	€80 000

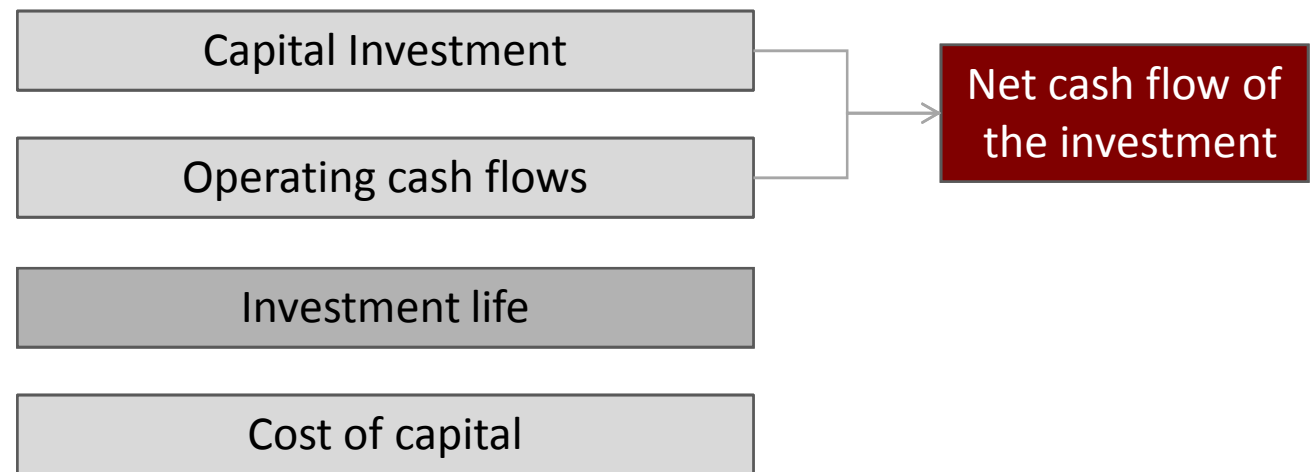
Year 2 Profit for year €20 000 (€40 000 less depreciation €20 000).

Balance sheet as at Year end 2

	Cost	Depreciation	Book value
Fixed asset	€100 000	€40 000	€60 000

Investment appraisal techniques highlight the key investment factors which are central to the profitability of an investment

They are:



All four are interrelated and eventually have to be looked at in unison

Most assets have a finite life and will eventually have to be scrapped or otherwise disposed of

The determinants of this life may be:

Physical

Wear and tear, or metal fatigue may cause the asset to collapse or be no longer serviceable

Technical

Assets may become obsolescent because a machine has come on to the market that is technically and economically superior to the existing one

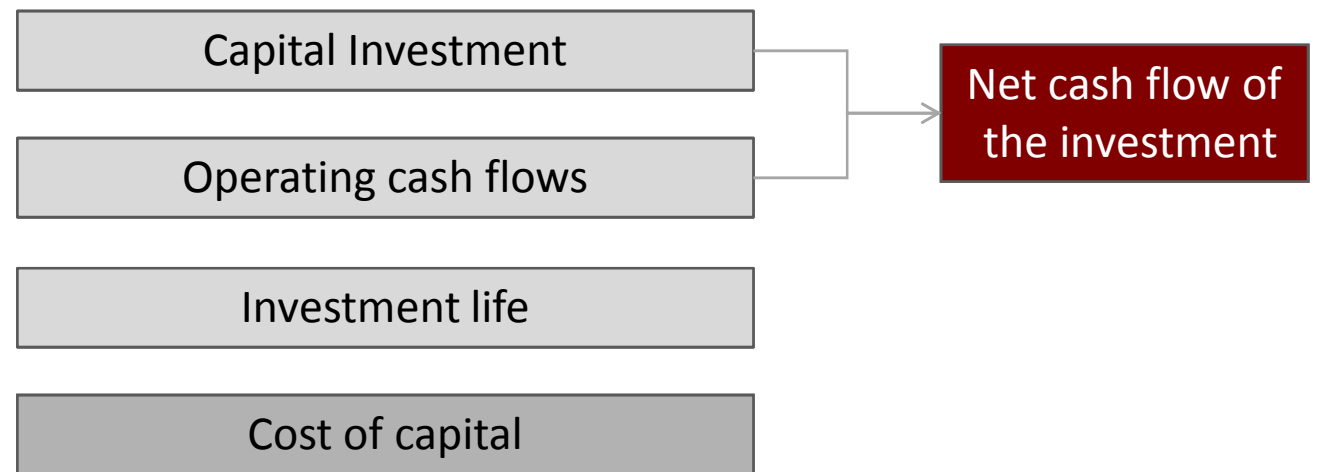
Market

The product or service which the asset provides may no longer be in demand in the marketplace

The investment life for appraisal purposes will be the estimate of which of the three factors is likely to occur first

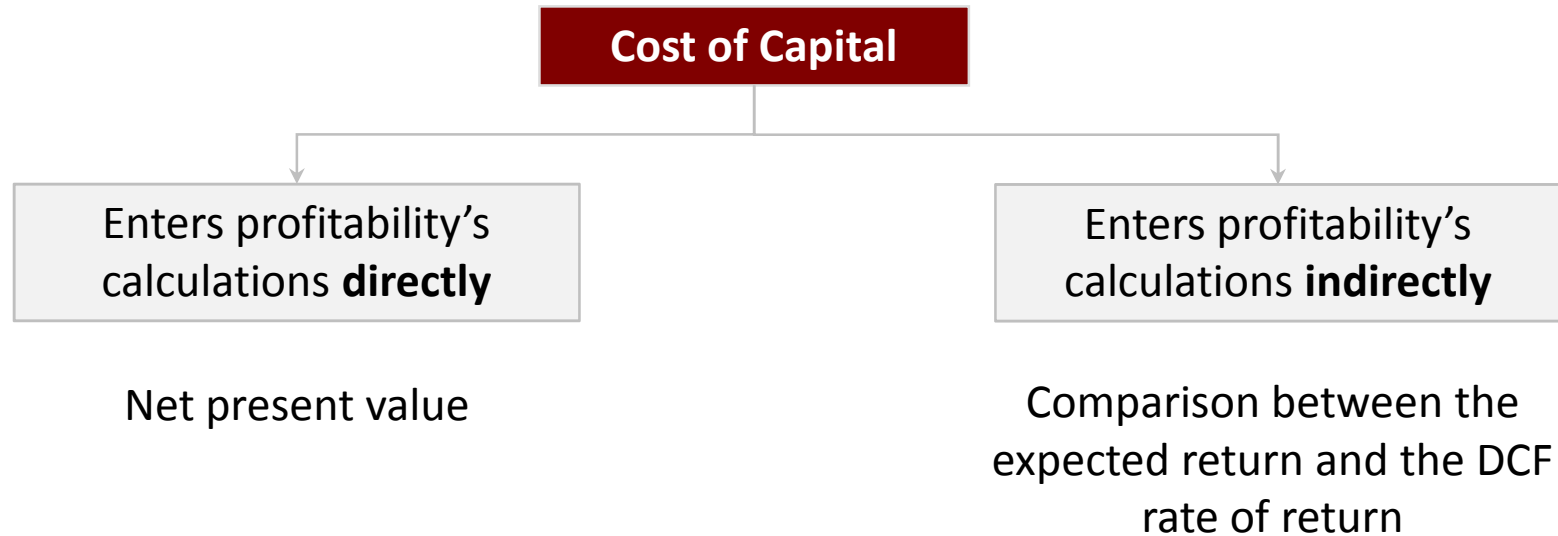
Investment appraisal techniques highlight the key investment factors which are central to the profitability of an investment

They are:



All four are interrelated and eventually have to be looked at in unison

The cost of capital is a key factor in assessing the profitability of a project



For most managers the cost of capital will be laid down by the board of directors or senior management and it will be the target the managers have to achieve

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

Cost of capital is regarded as the average cost of all different types of sources of long-term capital

Major Sources of Long-Term Finance

Fixed interest loans
(e.g. debentures)

Interest rate
+ Servicing Cost
- Taxation

Cost Of Capital

If the interest rate is 12% & the service cost is assumed to be 0.5% & the appropriate tax rate is 30%

Cost of Capital

$$\begin{aligned} &= (12\% + 0.5\%) \times (1 - 0.3) \\ &= 8.75\% \end{aligned}$$

Fixed interest (dividend) shares
(preference shares)

Net interest/ dividend
+ Cost of raising it

Cost Of Capital

If the interest or dividend is 10% & the service cost is again assumed to be 0.5% then the cost is 10.5%

Cost of Capital

$$\begin{aligned} &= 10\% + 0.5\% \\ &= 10.5\% \end{aligned}$$

In UK, there are no taxes on dividends

Cost of capital is regarded as the average cost of all different types of sources of long-term capital

Major Sources of Long-Term Finance (Continued)

Residual equity shares

E.g.: a new issue of ordinary shares, rights issues and other non-free issues of ordinary shares (No predetermined rate of interest or dividend)



There must be an imputed return which the company must seek to achieve & company should at least maintain its present value (with no dilution)

Retained earnings (Profits)

These are the earnings the business has generated internally through not distributing all the profits it earns



If the relative value of the shareholders' interest in the business has to be maintained, these funds should earn at least the same as the existing investment

Example showing how to calculate the target rate of return for residual equity shares or what we can call cost of capital

Example

Suppose a company has, at present, 100,000 ordinary €1 shares already issued and its current after-tax earnings are €20,000, while the market value of a €1 ordinary share is €2

$$\text{EPS} = \frac{\text{Net profit}}{\text{\# of ordinary shares in issue}} = \frac{€20,000}{100,000} = 20\text{p}$$

$$\text{Earnings Yield} = \frac{\text{EPS}}{\text{Market value / share}} \times 100 = \frac{20\text{p}}{€2} = 10\%$$

If there were no costs of issuing such shares and €2 was actually received for every share issued then the rate of return for the cost of capital calculation would be 10%

Example showing how to calculate the target rate of return for residual equity shares or what we can call cost of capital

Example (Continued)

However, there will be issue expenses and the cash actually received will be less than the price paid, so this will have to be allowed for. If issue costs are, say, 5% of cash paid, that is, out of the €2 the company only receives €1.90

Then the target rate of return will have to be higher than the 10% previously calculated

If €50,000 is to be raised by the issue of ordinary shares then the number of shares to be issued at a price of €2 will be:

$$\frac{€50,000}{€1.90} = 26,316$$

and to earn on that number of shares the same 20p per share as before the earnings will have to be $26,316 \times 20\text{p}$ which equals €5,263. The target return on the €50,000 invested will therefore be:

$$\frac{€5,263}{€50,000} = 10.5\%$$

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

After arriving at the estimated costs of the various sources of long-term finance, we then work out an average cost based on the relative proportions expected to be used

Example

Suppose the corporate plan shows a projected capital structure as follows:

Source of finance	Optimum proportion (%)	Estimated cost
Long-term loans	30	8.75%
Preference share capital	5	10.50%
Existing ordinary share capital	20	10.00%
New ordinary share capital	10	10.50%
Retained earnings	35	10.00%
	<u>100</u>	

Using the optimum proportions as weights, the average cost of capital can be calculated as:

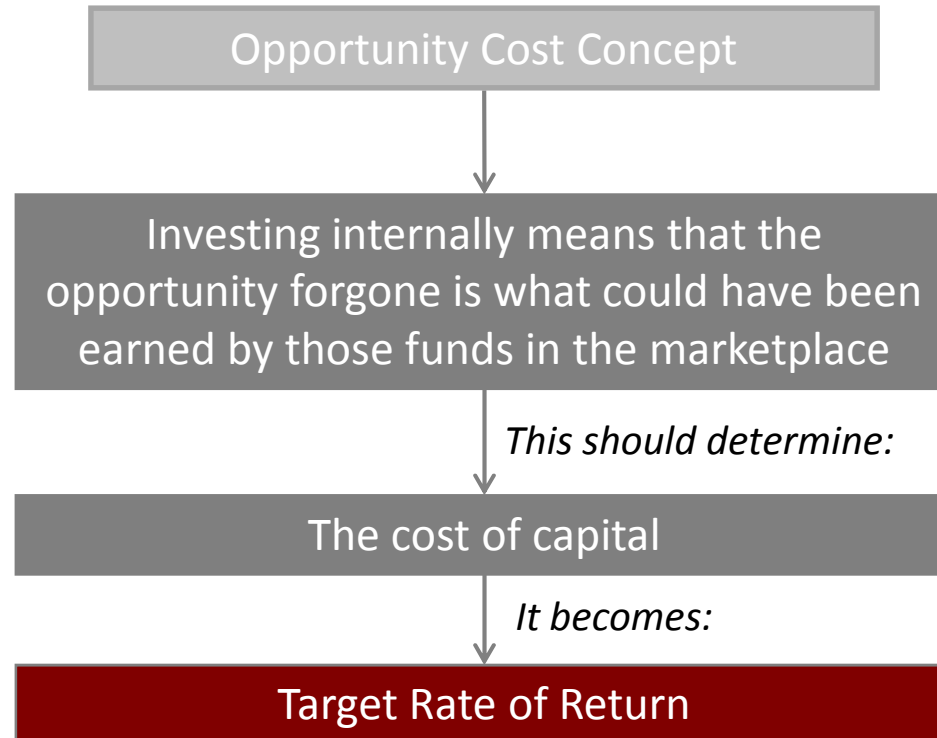
Long-term loans	$0.30 \times 8.75 = 2.625$	
Preference share capital	$0.05 \times 10.50 = 0.525$	
Existing ordinary share capital	$0.20 \times 10.00 = 2.000$	
New ordinary share capital	$0.10 \times 10.50 = 1.050$	
Retained earnings	$0.35 \times 10.00 = 3.500$	
	<u>9.700</u>	<i>Average Cost of Capital</i>

The analysis is further complicated by another two factors – opportunity cost and risk

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

Funds should only be used internally if the return achieved is at least as good as that which could be achieved by investing outside the organization



The cost of raising finance may set the minimum figure but management should be aiming to achieve something higher than this

Case

- The AB Company is considering investing €100,000 in capital equipment – payable €60,000 on receipt, €40,000 in one year. Installation costs €10,000. The machine will last for ten years at the end of which the scrap value will be €2,500. Working capital will increase as follows:

Debtors	€30 000
Creditors	20 000
Stocks	15 000

- The net operating cash flows will be €30,000 per annum for the first four years and thereafter €35,000 per annum.
- The cash flows do not include interest on an increased bank overdraft of €15,000 which is estimated at €2,500 per annum.
- The company intends issuing a ten-year €50,000 12% long-term loan (debenture) repayable at par.
- The company's cost of capital is based on a target long-term capital structure of:

60 per cent equity at cost of 18 per cent;
40 per cent loans (debentures) at average cost of 13 per cent.

Case (Continued)

- The cash profile of the proposed investment is therefore:

	Equipment	Installation	Working capital	Bank overdraft	Operating cash flows	Total cash flow
	€	€	€	€	€	€
Year 0	-60 000	-10 000		15 000		-55 000
1	-40 000		-25 000	-2 500	30 000	-37 500
2				-2 500	30 000	27 500
3				-2 500	30 000	27 500
4				-2 500	30 000	27 500
5				-2 500	35 000	32 500
6				-2 500	35 000	32 500
7				-2 500	35 000	32 500
8				-2 500	35 000	32 500
9				-2 500	35 000	32 500
10				-17 500	35 000	17 500
11	2500		25 000			27 500

Short-term loans do not form part of target capital structure and do not enter into the calculation of the cost of capital

Working capital investment takes place in Year 1, and is recovered at end of project

Case (Continued) Long term loans are unlike short term ones (Bank overdraft) as they are not included in the cash profile and form part of the target capital structure

Short term loans (*overdraft*)

Included in the cash profile

Do **NOT** form part of target capital structure

Do **NOT** enter into the calculation of the cost of capital

Long term loans (*debentures*)

Not included in the cash profile

Forms part of target capital structure

Enters into the calculation of the cost of capital



Enters in cost of capital calculations:

60% Equity at 18%	10.80%
40% Long-term loans at 13%	5.20%
	<u>16.00%</u>

If the project is now evaluated, it gives a net present value of €34,007 at 16% or a rate of return of 24.4%; It breaks even after 6.5 years

	Cash flow	PV 16% factor	PV cash flow at 16%	PV 24% factor	PV cash flow at 24%	PV 28% factor	PV cash flow at 28%
	€		€		€		€
Year 0	–55 000	1.000	–55 000	1.000	–55 000	1.000	–55 000
1	–37 500	0.862	–32 325	0.806	–30 225	0.781	–29 287
2	27 500	0.743	20 432	0.650	17 875	0.610	16 775
3	27 500	0.641	17 627	0.524	14 410	0.477	13 117
4	27 500	0.552	15 180	0.423	11 632	0.373	10 257
5	32 500	0.476	15 479	0.341	11 082	0.291	9 457
6	32 500	0.410	13 325	0.275	8 937	0.227	7 377
7	32 500	0.354	11 505	0.222	7 215	0.178	5 785
8	32 500	0.305	9 912	0.179	5 817	0.139	4 517
9	32 500	0.263	8 547	0.144	4 680	0.108	3 510
10	17 500	0.227	3 972	0.116	2 030	0.085	1 487
11	27 500	0.195	5 362	0.094	2 585	0.066	1 815
			<u>€34 007</u>		<u>€1 038</u>		<u>–€10 190</u>

$$\begin{aligned}\text{Rate of return} &= 24\% + \left[4\% \times \frac{1038}{1038 + 10\,190} \right] \\ &= 24.4\%\end{aligned}$$

If required, a sensitivity analysis could be done on the key variables

Agenda

- 15.1 Introduction
- 15.2 The Investment Process
- 15.3 Concept of Present Value
- 15.4 Discounted Cash Flow Approach
- 15.5 Net Present Value (NPV)
- 15.6 Discounted Cash Flow (DCF) Rate of Return
- 15.7 Comparison of Net Present Value and DCF Rate of Return
- 15.8 Investment Appraisal in Non-Revenue and Not-for-Profit Situations
- 15.9 Risk and Uncertainty and Inflation
- 15.10 Risk and Uncertainty
- 15.11 Payoff or Payback Period
- 15.12 Sensitivity Analysis
- 15.13 Risk Analysis
- 15.14 The Key Investment Factors
- 15.15 Projected Average Cost of Capital
- 15.16 Average Cost of Capital
- 15.17 Opportunity Cost, Risk and the Cost of Capital
- 15.18 Investment Appraisal and Inflation
- 15.19 Post-Assessment/Continuous Post-Audit of Capital Expenditure Projects

Inflation is important when evaluating differing cash flows over future periods and it should be incorporated into the evaluation in the same way as any other variable

Example

Selling price of
€10

3% Inflation

In 10 years
€13.43

In 10 years €10 will not represent the same purchasing power as today's...it will inc. by 34.3%

Different components of cost and revenues inflate (or deflate – occasionally prices do go down) at different rates



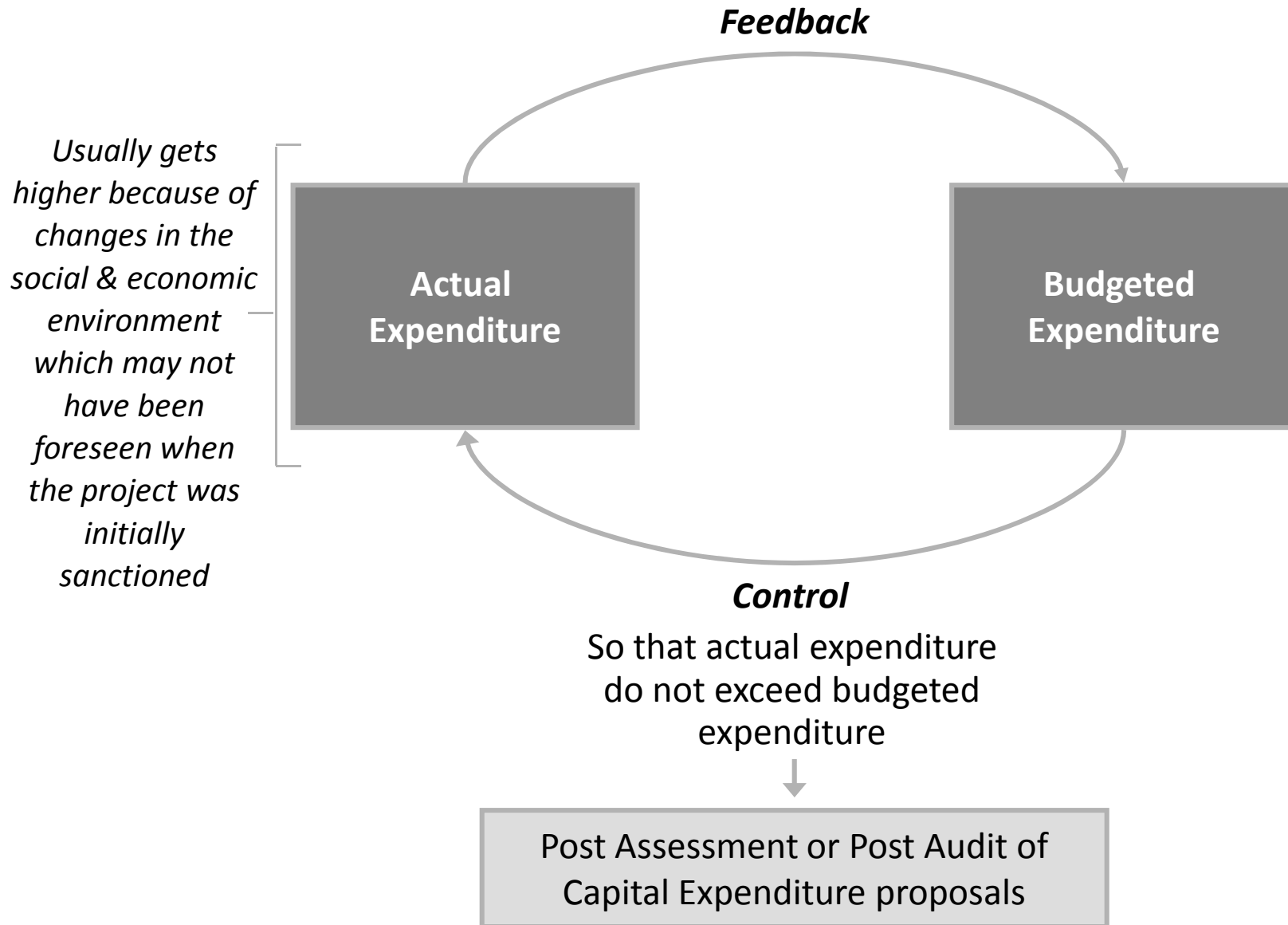
Therefore:

We *have to* build an allowance into our calculations for this effect separately from the time value of money

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- 15.10 Risk and Uncertainty
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- 15.14 The Key Investment Factors
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Once an investment has been undertaken, the financial control of it would follow (monitoring of actual income and expenditure against forecast)



Multiple Choice Questions (Continued)

15.1 Which of the following describes the concept of present value, as applied in the evaluation of investments?

- A. It takes account of the effects of inflation on the value of money.
- B. It makes allowance for the different risk factors attached to different projects.
- C. It is based on the ability to earn interest or the need to pay interest on investments.
- D. It is applied in circumstances where there are only estimates of future cash flows.

Multiple Choice Questions (Continued)

15.2 Discounted cash flow methods of investment appraisal split into two main techniques –net present value and accounting rate of return. True or false?

False

Multiple Choice Questions (Continued)

15.3 The use of the net present value (NPV) technique gives rise to some clear rules which can be applied in investment decisions. These are:

- i. Where NPV exceeds zero, the investment should be rejected.
- ii. Where NPV equals zero, the investment should be rejected.
- iii. Where NPV is less than zero, the investment should be rejected.
- iv. Where NPV exceeds zero, the investment should be accepted.

Which of the following is correct?

- A. (i) and (ii) only.
- B. (i), (ii) and (iii) only.
- C. (ii) and (iii) only.
- D. (iii) and (iv) only.

Multiple Choice Questions (Continued)

The following information applies to Questions 15.4–15.7.

Midland Engineering Limited is considering an investment of €20,000 in a new milling machine. The Production Manager has estimated that its acquisition will result in extra annual income of €7,000 over the next four years. The company's cost of capital is 8%.

15.4 What is the total present value of the income from this investment?

- A. +€3,184.
- B. +€22,771.
- C. +€23,184.
- D. +€23,597.

Year	Cash flow €	8% PV factor	PV at 8% €
0	–20 000	1.000	–20 000
1	+7 000	0.926	+6 482
2	+7 000	0.857	+5 999
3	+7 000	0.794	+5 558
4	+7 000	0.735	+5 145
			<u>+3 184</u>

The total PV of the income stream is €23 184, before the deduction of the investment of €20 000 cuts it to a NPV of €3184.

Multiple Choice Questions (Continued)

15.5 What is the net present value of this investment?

A. +€2,771.

B. +€3,184.

C. +€3,597.

D. +€23,184.

Multiple Choice Questions (Continued)

15.6 If the company's cost of capital increased to 10 per cent, what is the net present value of this investment?

- A. +€2,183.
- B. -€2,183.
- C. +€2,659.
- D. +€22,183.

Year	Cash flow €	10% PV factor	PV at 10% €
0	-20 000	1.000	-20 000
1	+7 000	0.909	+6 363
2	+7 000	0.826	+5 782
3	+7 000	0.751	+5 257
4	+7 000	0.683	+4 781
			+2 183

Multiple Choice Questions (Continued)

15.7 If the extra annual income declined by 10%, with the cost of capital remaining at 8%, what is the net present value of this investment?

A. –€866.

B. +€866.

C. +€1,237.

D. +€1,730.

Year	Cash flow €	8% PV factor	PV at 8% €
0	–20 000	1.000	–20 000
1	+6 300	0.926	+5 834
2	+6 300	0.857	+5 399
3	+6 300	0.794	+5 002
4	+6 300	0.735	+4 631
			+866

Multiple Choice Questions (Continued)

15.8 An important feature of the NPV technique is its focus on applying discounting factors only to capital expenditure. True or false?

False

Multiple Choice Questions (Continued)

15.9 The use of the internal rate or return (IRR) technique gives rise to some clear rules which can be applied in investment decisions. These are:

- i. Where IRR exceeds the cost of capital, the investment should be accepted.
- ii. Where IRR equals the cost of capital, the investment should be rejected.
- iii. Where IRR is less than the cost of capital, the investment should be rejected.
- iv. Where IRR exceeds the cost of capital, the investment should be rejected.

Which of the following is correct?

- A. (i) and (ii) only.
- B. (i), (ii) and (iii) only.
- C. (i) and (iii) only.
- D. (ii), (iii) and (iv) only.

Multiple Choice Questions (Continued)

The following information applies to Questions 15.10–15.11.

Southland Devices Limited is considering the investment of €80,000 in a new clean room facility for the manufacture of integrated circuits. It is estimated that this investment will generate extra revenue of €25,000 per annum in the first three years, falling to €20,000 per annum in the final year of its estimated lifetime.

15.10 What is the DCF rate of return on this investment?

Try 8 per cent and then try 6 per cent.

A. 6.50%

B. 6.75%

C. 7.12%

D. 7.51%

Year	Cash flow €	8% PV factor	PV at 8% €	6% PV factor	PV at 6% €
0	–80 000	1.000	–80 000	1.000	–80 000
2	+25 000	0.857	+21 425	0.890	+22 250
3	+25 000	0.794	+19 850	0.840	+21 000
4	+20 000	0.735	+14 700	0.792	+15 840
			<u>–875</u>		<u>+2 665</u>

Interpolate to arrive at correct DCF rate:

$$= 6\% + \left(\frac{2665}{2665 + 875} \times 2\% \right) = 7.51\%$$

Multiple Choice Questions (Continued)

15.11 If the company's cost of capital is 6.5%, which of the following should represent the company's final assessment of the investment?

- A. Proceed with investment: cost of capital exceeds IRR.
- B. Reject the investment: cost of capital exceeds IRR.
- C. Reject the investment: IRR exceeds cost of capital.
- D. Proceed with investment: IRR exceeds cost of capital.

Multiple Choice Questions (Continued)

15.12 Where a company is faced with a choice between two mutually exclusive investment options, the application of either the NPV or the DCF rate of return technique will always result in an identical choice of investment. True or false?

False

Multiple Choice Questions (Continued)

15.13 The DCF rate of return is that discount rate which, when applied to the future cash flows, results in a net present value of exactly zero. True or false?

True

Multiple Choice Questions (Continued)

The following information applies to Questions 15.14–15.15.

Plastic Models Limited is faced with a pressing requirement to introduce a computer system for use in its manufacturing control, order processing and accounting systems. The Financial Director has been investigating the available options and has narrowed the choice down to two computer systems, both of which will satisfy the technical requirements of the company.

The Easilog II computer can be acquired for €12,000, but is expected to incur, over its four-year lifetime, costs of €4,000 in the first three years and €2,000 in the final year. The other acceptable computer system, the Peach computer, will cost €10,000 to purchase, but will have operating costs over its five-year lifetime of €3,000 in both Year 1 and Year 2, €2,000 in both Year 3 and Year 4, and €1,000 in the final year.

Multiple Choice Questions (Continued)

15.14 If the company's cost of capital is 12%, what is the NPV of the Easilog II computer?

- A. €1,120.
- B. €10,880.
- C. €21,596.
- D. €22,880.

Year	Cash outflow €	12% PV factor	PV at 12% €
0	12 000	1.000	12 000
1	4 000	0.893	3 572
2	4 000	0.797	3 188
3	4 000	0.712	2 848
4	2 000	0.636	1 272
			22 880

Multiple Choice Questions (Continued)

15.15 In selecting between the two systems, which of the following is correct?

- A. The company should select the Peach computer, since it has the lower NPV.
- B. The company should select the Peach computer, since it has the higher NPV.
- C. The company should select the Easilog II computer, since it has the higher NPV.
- D. The company should select the Easilog II computer, since it has the lower NPV.

Year	Cash outflow €	12% PV factor	PV at 12% €
0	10 000	1.000	10 000
1	3 000	0.893	2 679
2	3 000	0.797	2 391
3	2 000	0.712	1 424
4	2 000	0.636	1 272
5	1 000	0.567	567
			18 333

Since the decision refers to a cost minimisation (rather than income maximisation) exercise, then the correct option is to select the computer with the lower NPV.

Multiple Choice Questions (Continued)

15.16 In utilizing the DCF approach to investment appraisal, managers can be completely confident about the levels of risk and uncertainty involved in each and every investment option. True or false?

False

Multiple Choice Questions (Continued)

15.17 In tackling the issues of risk and uncertainty in investment evaluation, there is a variety of possible approaches, including:

- i. the use of a 'low hurdle' test for more risky projects;
- ii. the introduction of the payback period assessment;
- iii. the use of a 'high-hurdle' test for more risky projects;
- iv. the contribution margin technique.

Which of the following is correct?

- A. (i) and (ii) only.
- B. (ii) and (iii) only.
- C. (ii) and (iv) only.
- D. (iii) and (iv) only.

Multiple Choice Questions (Continued)

The following information applies to Questions 15.18–15.20.

Tyler and Moore Limited have set aside funds in their capital expenditure budget for the renewal of the company's packaging line. They have received a quotation from Automatic Packaging Limited which has put forward two of its products for consideration, the Auto-moderne and the Auto-pack. The details of each product are as follows:

	Auto-moderne	Auto-pack
	€	€
Purchase price	15 000	11 000
Annual cost savings		
Year 1	6 000	3 000
2	4 500	5 000
3	3 000	4 000
4	4 000	3 500

Multiple Choice Questions (Continued)

15.18 What is the payback period for the Auto-moderne product?

- A. 3.38 years.
- B. 3.50 years.
- C. 3.86 years.
- D. 4.00 years.

Year	Cumulative cash flow	
	€	
0	–€15 000	
1	–€9 000	(–€15 000 + €6 000)
2	–€4 500	(–€9 000 + €4 500)
3	–€1 500	(–€4 500 + €3 000)
4	+€2 500	(–€1 500 + €4 000)

Payback will occur during the fourth year, at the point at which the negative cumulative cash flow of €1500 (at the end of Year 3) is extinguished. Since we assume that the cash inflow of €4000 accrues evenly over the year, payback will occur in 3.38 years ($3 + 1500/4000$).

Multiple Choice Questions (Continued)

15.19 What is the payback period for the Auto-pack product?

- A. 1.89 years.
- B. 2.75 years.
- C. 3.00 years.
- D. 3.38 years.

Year	Cumulative cash flow	
	€	
0	–€15 000	
1	–€9 000	(–€15 000 + €6 000)
2	–€4 500	(–€9 000 + €4 500)
3	–€1 500	(–€4 500 + €3 000)
4	+€2 500	(–€1 500 + €4 000)

Payback will occur during the fourth year, at the point at which the negative cumulative cash flow of €1500 (at the end of Year 3) is extinguished. Since we assume that the cash inflow of €4000 accrues evenly over the year, payback will occur in 3.38 years ($3 + 1500/4000$).

Multiple Choice Questions (Continued)

15.20 Automatic Packaging Limited is anxious to shift some slow-moving stocks of the Automoderne product and decides to offer a €3,000 reduction in price, causing the management of Tyler and Moore Limited to revise its evaluation of the two products according to the payback method of assessment. Which of the following is correct?

- A. The management should now switch to the Auto-moderne, since it has a reduced payback period.
- B. The management should now switch to the Auto-pack, since it has a lower payback period.
- C. The management should now switch to the Auto-moderne, since it has a longer payback period.
- D. The management should now switch to the Auto-pack, since it has a longer payback period.

Case Study 15.1

The Board of Directors of York Technology Products Limited is reviewing its investment plans for the forthcoming financial year. At its next meeting, it is scheduled to discuss two proposals – the first prepared by the Technical Director and the second by the Sales Director.

Case Study 15.1 (Continued)

Project A – Orlando Devices Inc.

The Technical Director has been contacted by an American company – Orlando Devices Inc., which is prepared to offer York the opportunity to manufacture a patented product under license, for exclusive sale in the Western European market. The new product – codenamed Tempus – will complement York's existing product range and no additional fixed manufacturing costs will be required before production can be commenced. Sales and service support can be carried out by York's current staff. However, additional marketing expenditure will be needed to launch the product and to maintain its identity in the marketplace. An initial licensing fee of €100,000 will be payable to Orlando Devices Inc., together with an annual royalty of 4% of sales value. Projections have been prepared for a three-year period:

	Year 1	Year 2	Year 3
	€000	€000	€000
Sales	320	470	540
Variable manufacturing costs	160	240	270
Contribution	160	230	270
Additional marketing costs	100	120	130
Net contribution	60	110	140

In order to provide adequate customer service and delivery, Orlando Devices Inc. insists that a minimum of two months' sales should be held in inventory at all times.

Case Study 15.1 (Continued)

Project B – Nagisoya Inc.

The Sales Director has been involved in extensive discussions with Nagisoya Inc. of Korea with a view to York's taking a role as the exclusive distributor for its products within the EEC. Nagisoya is prepared to offer an initial contract for a three-year period, but demands that York make a considerable investment to demonstrate its commitment to the success of the project.

In particular, Nagisoya's conditions for granting this potentially lucrative contract are that:

- (a) York agrees to a minimum annual purchase commitment of 500 units per annum, at an agreed price of €450 each, payable on delivery;
- (b) York invests in a properly equipped service workshop to support the product line. The estimated cost of equipping the workshop (in existing premises) is €75,000, with annual operating costs of €25,000. If the contract terminates at the end of Year 3, there will be no residual value in the equipment.

Case Study 15.1 (Continued)

Project B – Nagisoya Inc. (Continued)

The Sales Director has prepared the following forecast for the three-year period:

	Year 1	Year 2	Year 3
Sales (units)	400	900	1 400
Sales (price per unit)	<u>€675</u>	<u>€695</u>	<u>€720</u>
Other costs	€	€	€
Advertising	30 000	50 000	75 000
Salaries	20 000	35 000	63 000
Travel	5 000	14 000	33 000
Miscellaneous	15 000	23 000	18 000

Case Study 15.1 (Continued)

Currently, York has a capital structure which consists of:

	%	Cost per annum
Ordinary share capital	25	10.00%
Preference share capital	15	8.50%
Retained profits	45	10.00%
Debenture stock	15	4.83%
	100	

Required

The Managing Director has asked you, as Finance Director, to evaluate these proposals. Although both seem attractive, and there is no shortage of investment funds, he considers that only one may be effectively progressed at this time. Prepare your assessment of these projects, ignoring any taxation consequences.

Case Study 15.1 (Solutions)

Average cost of capital

The first step is to determine the relative percentage discount factor to apply to the future cash flows, that is, to compute the average cost of capital:

			%
Ordinary share capital	0.25 ×	10.00	2.500
Preference share capital	0.15 ×	8.50	1.275
Retained profits	0.45 ×	10.00	4.500
Debenture stock	0.15 ×	4.83	0.725
Average cost of capital			<u>9.000</u>

Case Study 15.1 (Solutions)

					4% of €320,000	4% of €470,000	4% of €540,000
Project A – Orlando Devices Inc.							
	Year 0	Year 1	Year 2	Year 3			
	€	€	€	€			
Licence fee	-100 000	-	-	-			
Net contribution	-	+60 000	+110 000	+140 000			
Royalty	-	-12 800 (1)	-18 800 (1)	-21 600 (1)			
Inventory	-	-40 000 (2)	-5 000 (2)	+45 000 (2)			
	-100 000	+7 200	+86 200	+163 400			
9% PV factor	1.00	0.917	0.842	0.772			
PV at 9%	-100 000	+6 602	+72 580	+126 145			
NPV = +€105 327							
					1/6 of €240,000	1/6 of €270,000 - €40,000	Inventory level is assumed to get reduced to zero

Case Study 15.1 (Solutions)

Project B – Nagisoya Inc.

	Year 0	Year 1	Year 2	Year 3
	€	€	€	€
Workshop	-75 000	-	-	-
Operating costs	-	-25 000	-25 000	-25 000
Sales	-	+270 000 (1)	+625 500 (1)	1 008 000 (1)
Purchases	-	-225 000 (2)	-360 000 (2)	-630 000 (2)
Other costs	-	-70 000	-122 000	-189 000
	-75 000	-50 000	+118 500	+164 000
9% PV factor	1.00	0.917	0.842	0.772
PV at 9%	-75 000	-45 850	+99 777	+126 608
NPV =	€105 535			

$400 \times \text{€}675$
 $900 \times \text{€}695$
 $720 \times \text{€}1400$
 $500 \times \text{€}450$
 $800 \times \text{€}450$
 $1,400 \times \text{€}450$

Case Study 15.1 (Solutions)

Summary

On the basis of the foregoing assessment and given that there is no shortage of investment funds, Project B is to be preferred to Project A since it generates a higher NPV for the company.

However, it may be appropriate to examine both investments from their payback perspective (before discounting).

		Project A	Project B
		Orlando Devices Inc.	Nagisoya Inc.
		€	€
Year	0	–100 000	–75 000
	1	–92 800	–125 000
	2	–6 600	–6 500
	3	+156 800	+157 500
Payback period		2.04 years	2.04 years

In this case, there is no difference in payback periods, leaving the original conclusion intact